CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Kameneva

Generated on 2026-10-16 14:09

Run

Split	train
Stimulus penalisation	0.2
Prior penalisation	1.0
Genes	97
Stimuli	1
Observed cells	1260
Timepoints	0, 8, 32, 48, 64, 80
Cell types	precursors, intermediate, sympathoblasts, chromaffin
Network shown	inter_simul.npy
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (observed / depth factor, model draws at the reference depth)

Perturbations (perturbation_simulation)

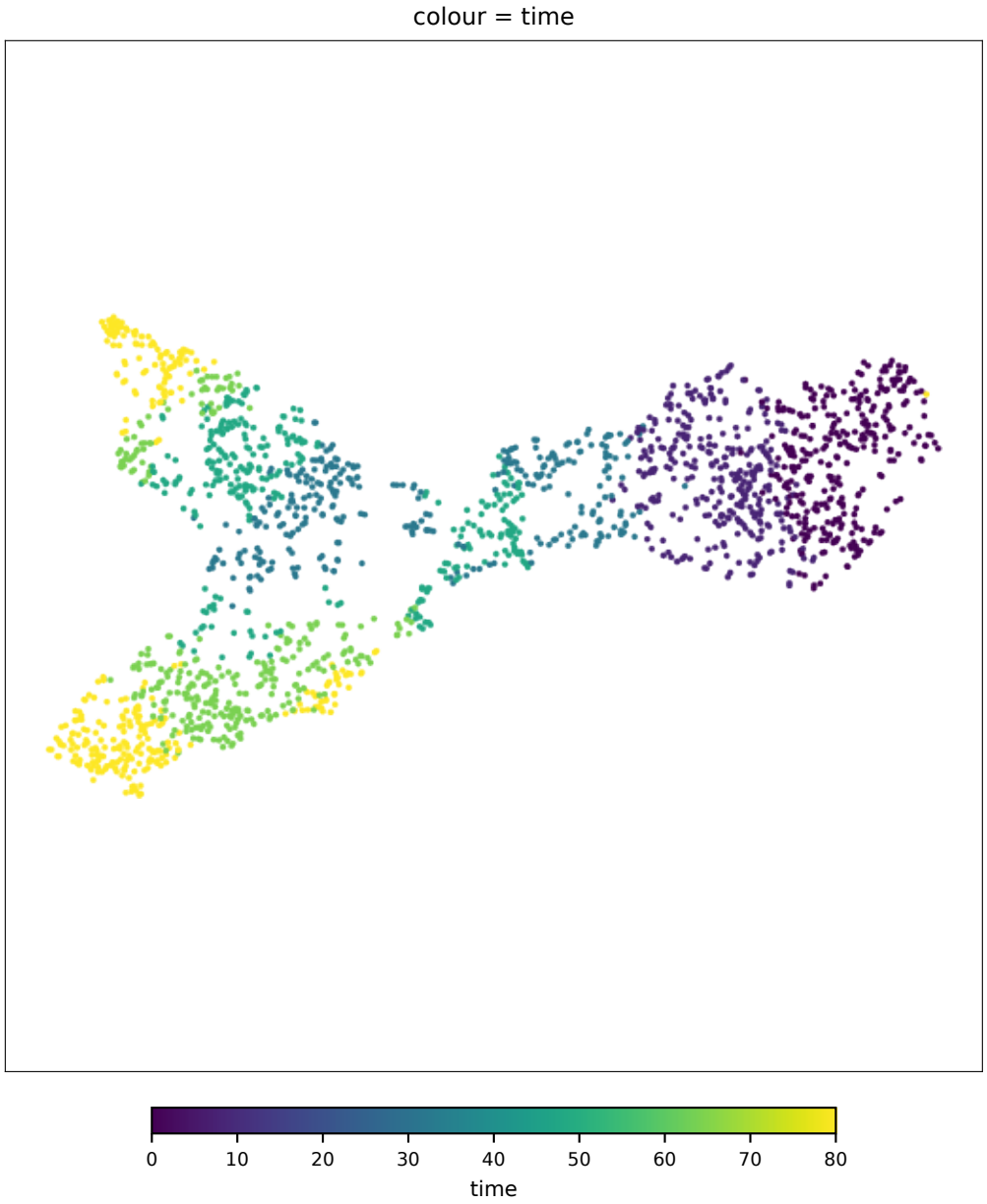
- ✓ P1 KO CHGA (50%)
- ✓ P2 KO CHGA
- ✓ P3 OV STMN2
- ✓ P4 KO CHGA · OV STMN2

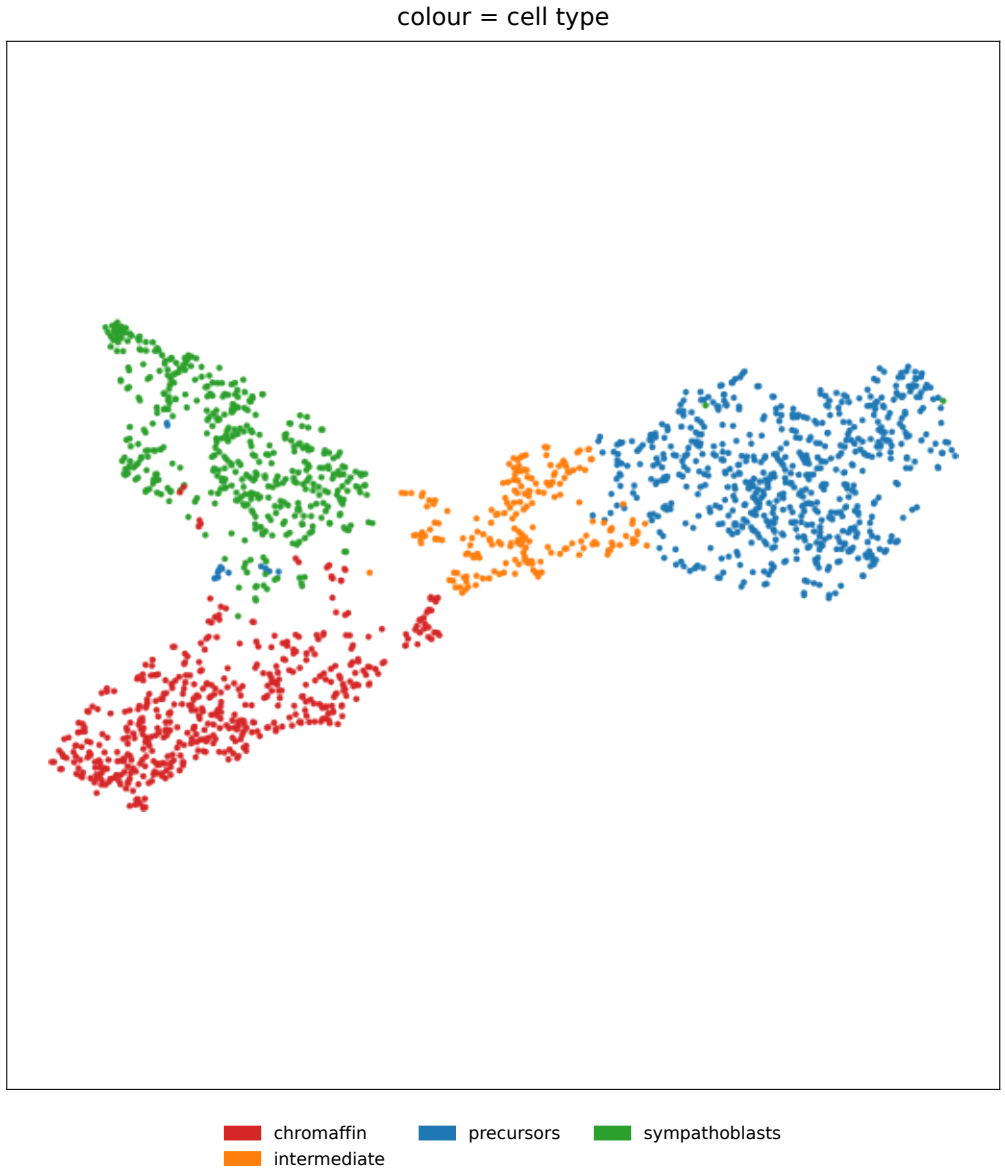
Contents

Data — UMAP of all the cells of Data/data.h5ad (time, samples); teaser: classical OT vs CardamomOT displacement fields (all times, then per interval), then cell-type transitio
1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots), top-20 regulators and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

Data — UMAP of all the cells

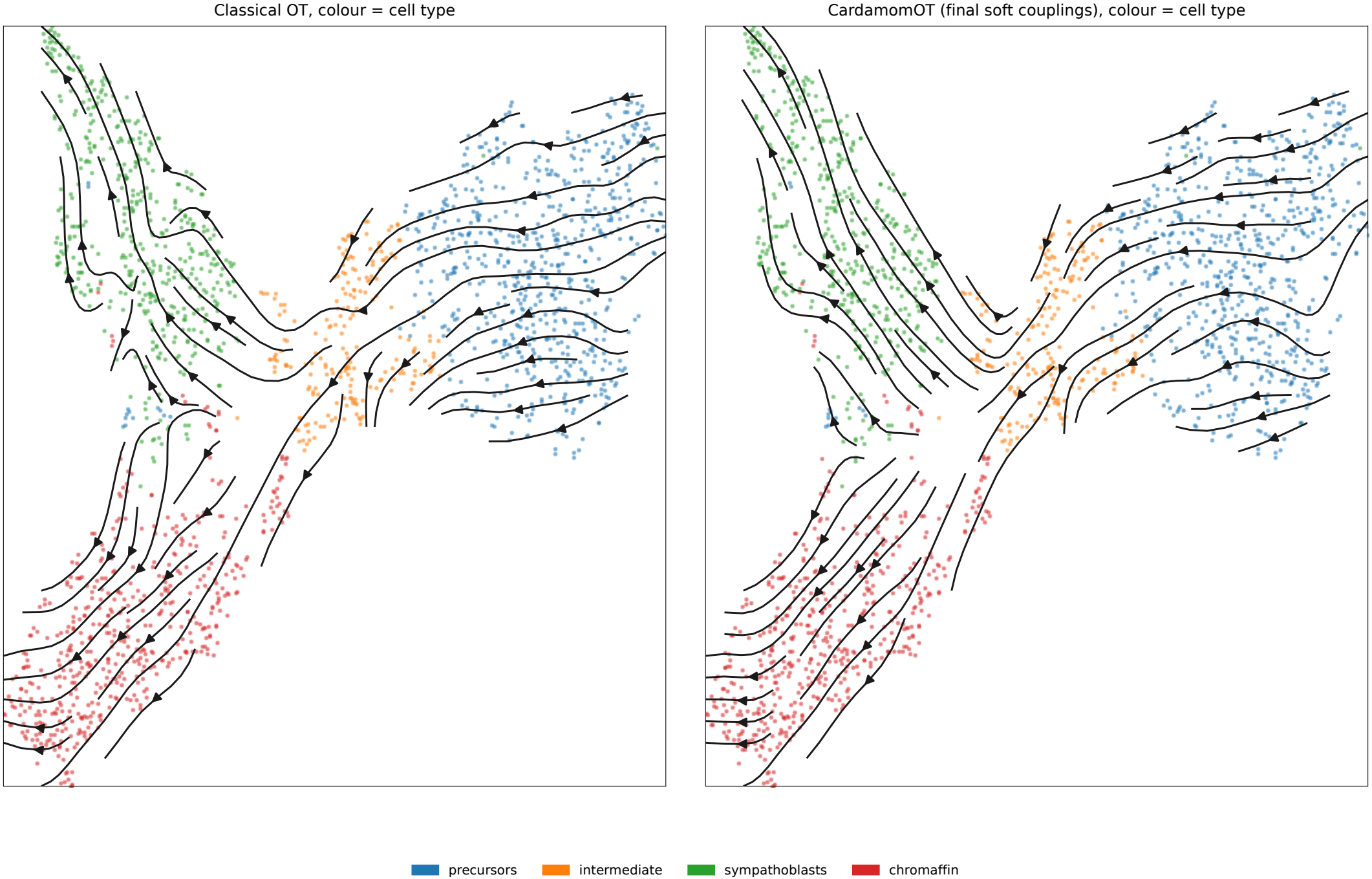
Data/data.h5ad, 1800 cells: log1p(counts), 97 highly variable genes, PCA (50 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.





Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.68 (PCA space). CardamomOT: final soft couplings of the trajectories (1260 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



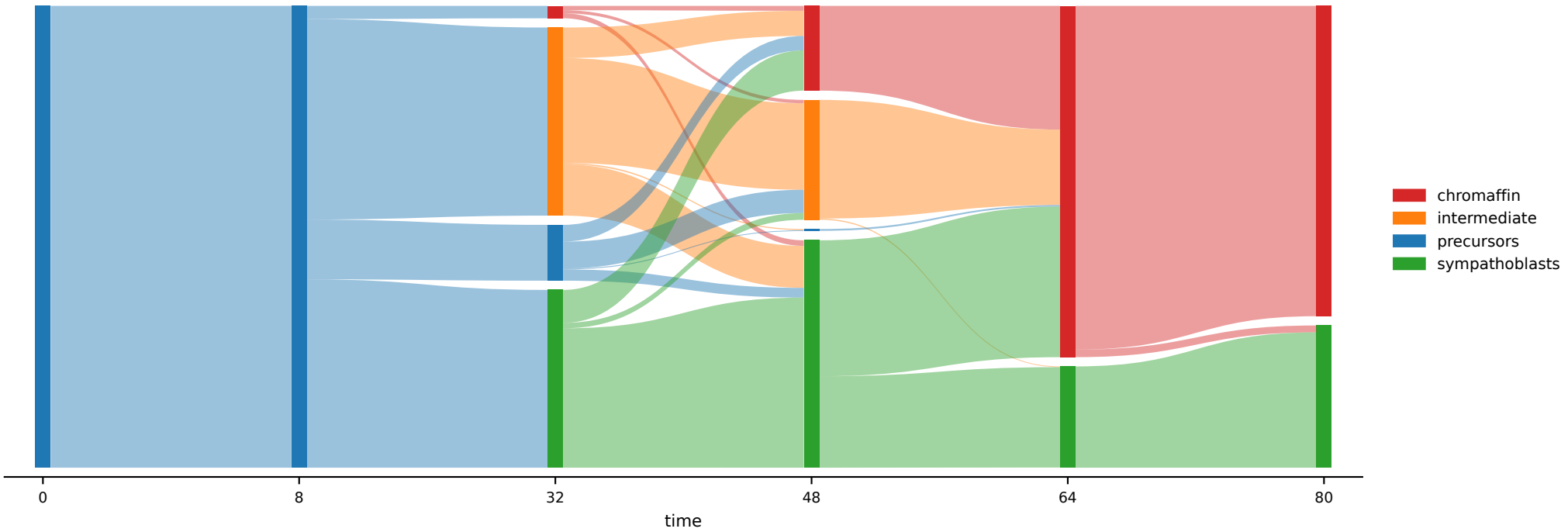
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) towards their mean descendant at t_to, kNN projection among the cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

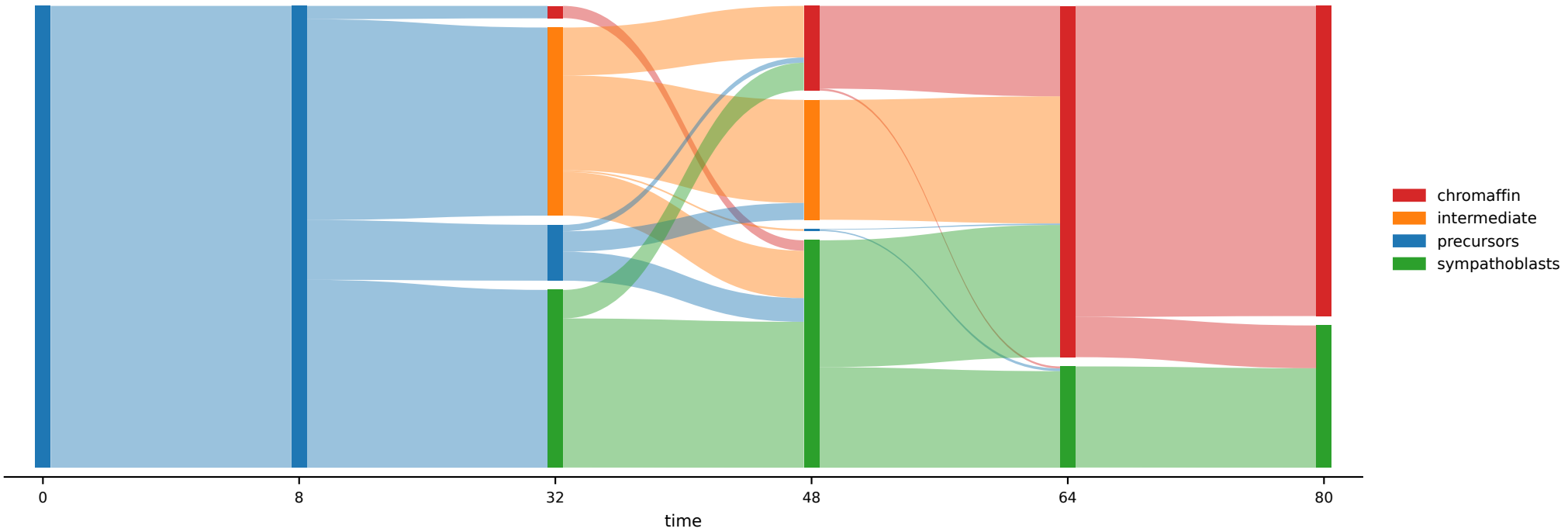
Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
chromaffin	**CHGA** **CHGB** MEG3 **DLK1** **PCSK1N** **TH** **SCG2** **RAMP1** **DBH** RGS4	**CHGA** **DLK1** **CHGB** **RAMP1** **TH** **SCG2** HTATSF1 **PCSK1N** PENK **DBH**
intermediate	**PLP1** PTPRZ1 **ZEB2** **POSTN** **COL5A2** QKI EDNRB SPARC VIM MPZ	MT-CYB **POSTN** **ZEB2** COL1A1 **COL5A2** VCAN ALDH1A1 **PLP1** OLFML2A NRXN1
precursors	**S100B** **VIM** **METRN** S100A10 **PLP1** **SPARC** ANXA2 **EDNRB** **MPZ** LGALS1	**PLP1** **S100B** **VIM** **SPARC** **EDNRB** PTPRZ1 **MPZ** **METRN** COL5A2 IFITM3
sympathoblasts	PTMA **GAP43** **TMSB10** **STMN2** **RTN1** **MAP1B** **BASP1** **GAL** **UCHL1** MDK	**STMN2** **RTN1** **BASP1** **GAP43** **MAP1B** **GAL** PHOX2A **UCHL1** CD24 **TMSB10**

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
chromaffin	**CHGB** **CHGA** **TH** **DBH** **SCG2** **RAMP1** **PCSK1N** RGS4 RGS5 **MEG3**	**CHGA** DLK1 **CHGB** **TH** **MEG3** **RAMP1** **SCG2** HTATSF1 **DBH** **PCSK1N**
intermediate	**PLP1** **COL5A2** **POSTN** **ZEB2** EDNRB **PTPRZ1** QKI **OLFML2A** SPARC **COL1A1**	**COL1A1** **POSTN** **ZEB2** VCAN **COL5A2** NEFL **OLFML2A** ALDH1A1 **PTPRZ1** **PLP1**
precursors	**S100B** METRN **VIM** S100A10 **PLP1** ANXA2 **SPARC** **EDNRB** **MPZ** IFITM3	**PLP1** PTPRZ1 **SPARC** **EDNRB** ZEB2 **VIM** COL5A2 **S100B** **MPZ** QKI
sympathoblasts	**PTMA** **PHOX2A** **MAP1B** **RTN1** **GAP43** **STMN2** **CD24** **GAL** HAND2 **BASP1**	**STMN2** **RTN1** **MAP1B** **GAL** **BASP1** **GAP43** **PHOX2A** **CD24** **PTMA** UCHL1

Genes of the model (97)

Data/data_train.h5ad, alphabetical order.

ACTG1	EIF4A2	NEFL	RPL41
ALDH1A1	FAU	NPY	RPS13
ANXA2	FOS	NRXN1	RPS2
APOE	GAL	OLFML2A	RPS29
ATP5F1E	GAP43	PCSK1N	RPS4X
B2M	GAPDH	PENK	RTN1
BASP1	H2AFZ	PHOX2A	S100A10
BEX1	HAND2	PLP1	S100A6
BTF3	HAND2-AS1	PNMT	S100B
C4orf48	HINT1	POSTN	SCG2
CARTPT	HMGB2	PPIA	SPARC
CD24	HMGN2	PRPH	STMN2
CHGA	HNRNPA1	PTMA	TH
CHGB	HTATSF1	PTMS	TIMP3
CNTN1	IFITM3	PTN	TMSB10
COL18A1	IGFBP2	PTPRZ1	TPM1
COL1A1	ITM2B	QKI	UBC
COL1A2	LGALS1	RAMP1	UCHL1
COL5A2	MAP1B	RGS4	VCAN
CRYAB	MDK	RGS5	VIM
DBH	MEG3	RPL10	YBX1
DLK1	MEG8	RPL12	ZEB2
EDNRB	METRN	RPL30	
EEF1A2	MPZ	RPL32	
EEF2	MT-CYB	RPL34	

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) when the depth factor s_i is applied.

Depth

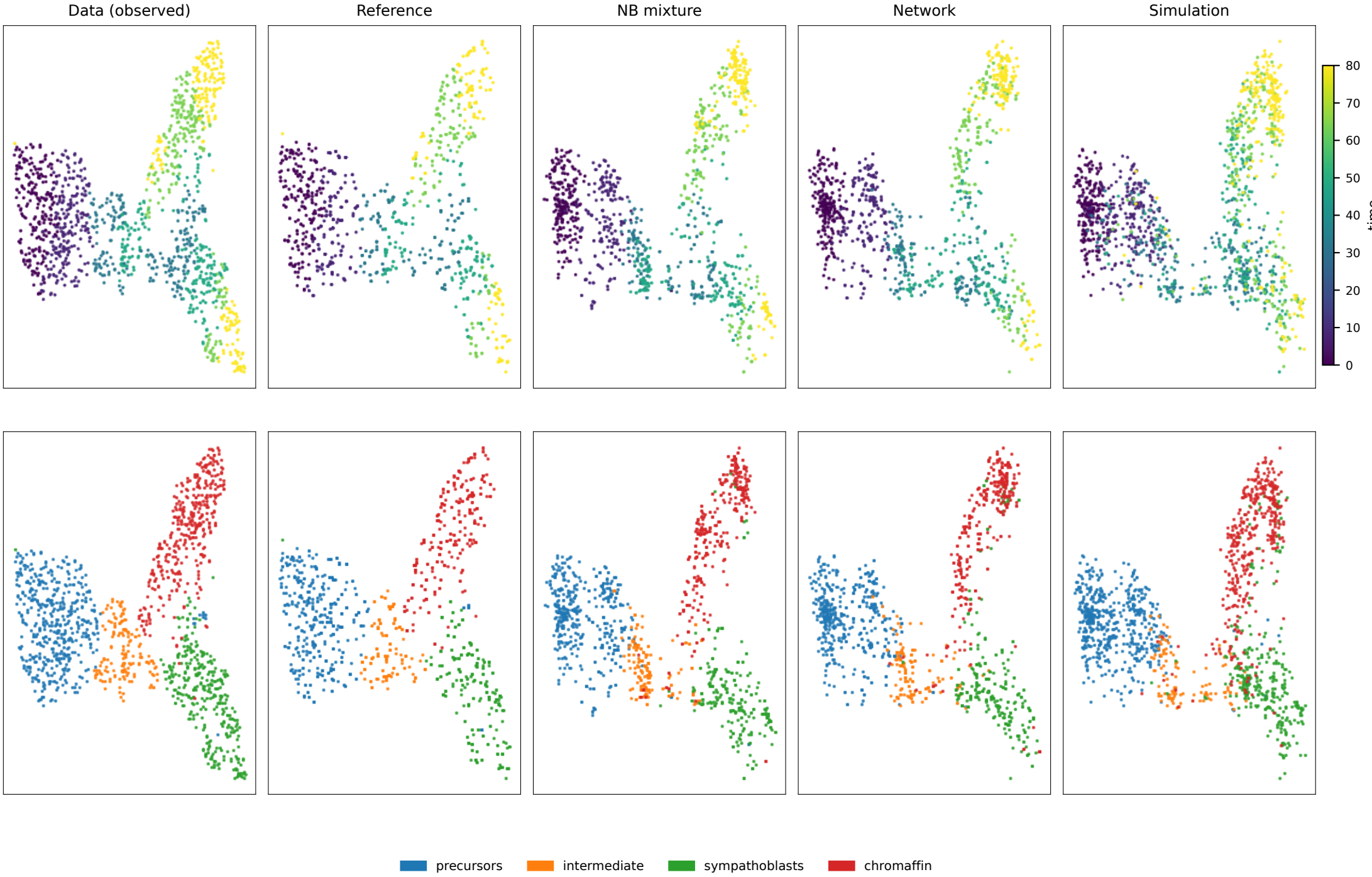
not estimated

Reason

97 genes < 10000

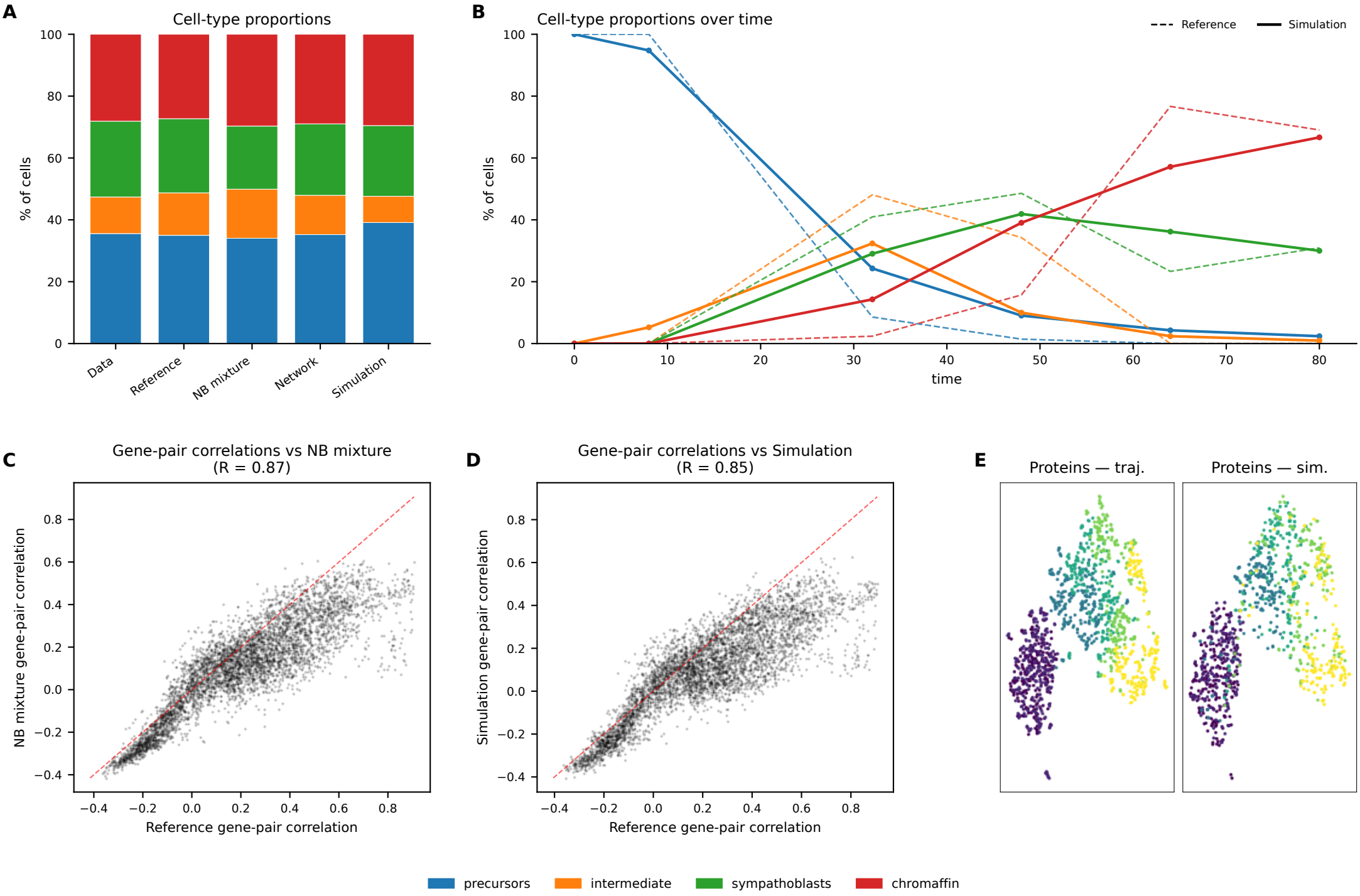
1. Generative model — trajectories and simulation

UMAP learned on the observed data; the trajectories (Reference: one state per ancestor, i.e. growth-weighted by exp ∫R_opt, as the simulation has proliferation), the NB mixture and network-driven modes along them and the full simulation are projected onto it.



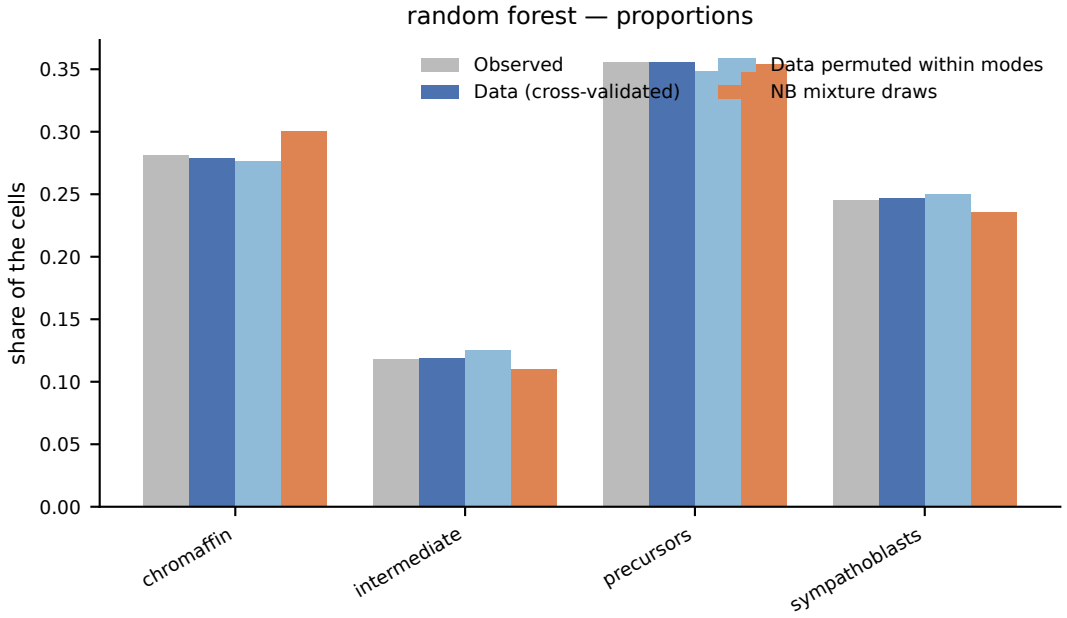
1. Generative model — quantitative checks

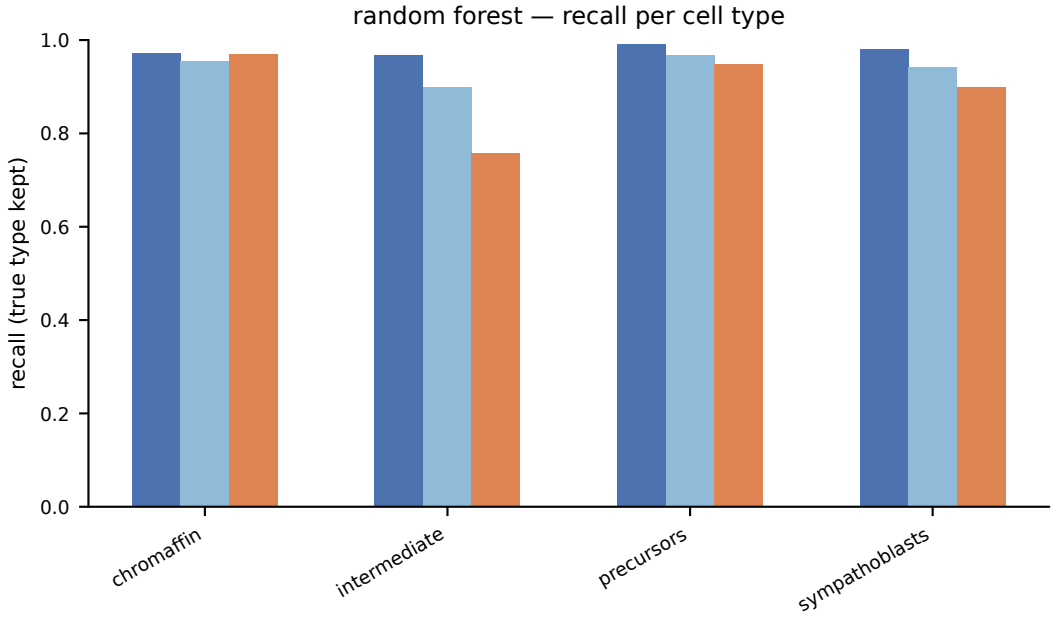
Reference = RNA of the trajectories equivalent to the simulation (growth-weighted: simulation with proliferation).

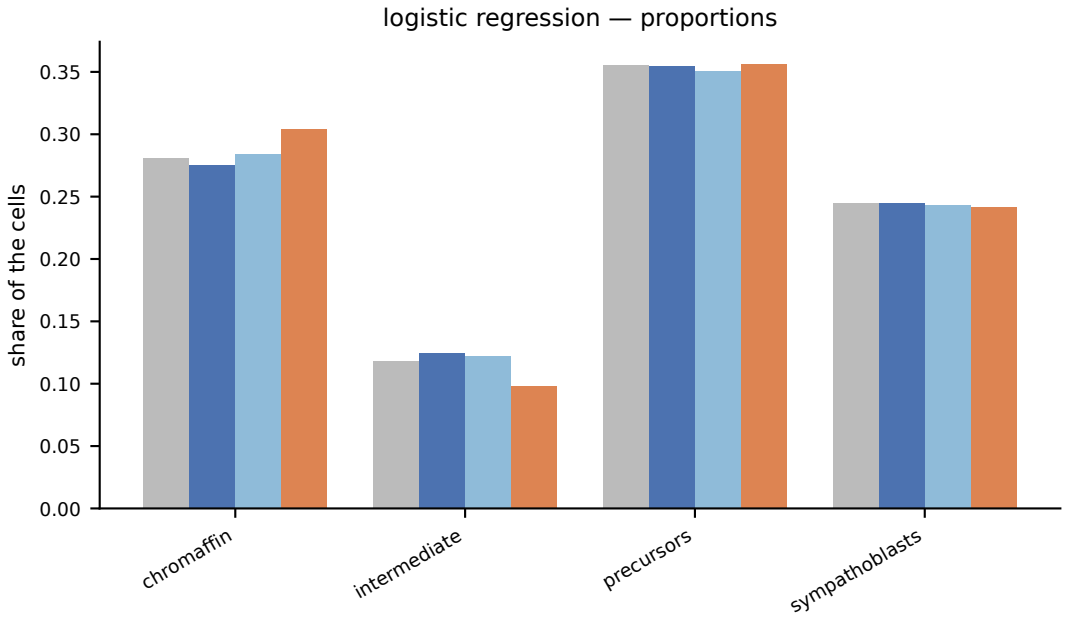


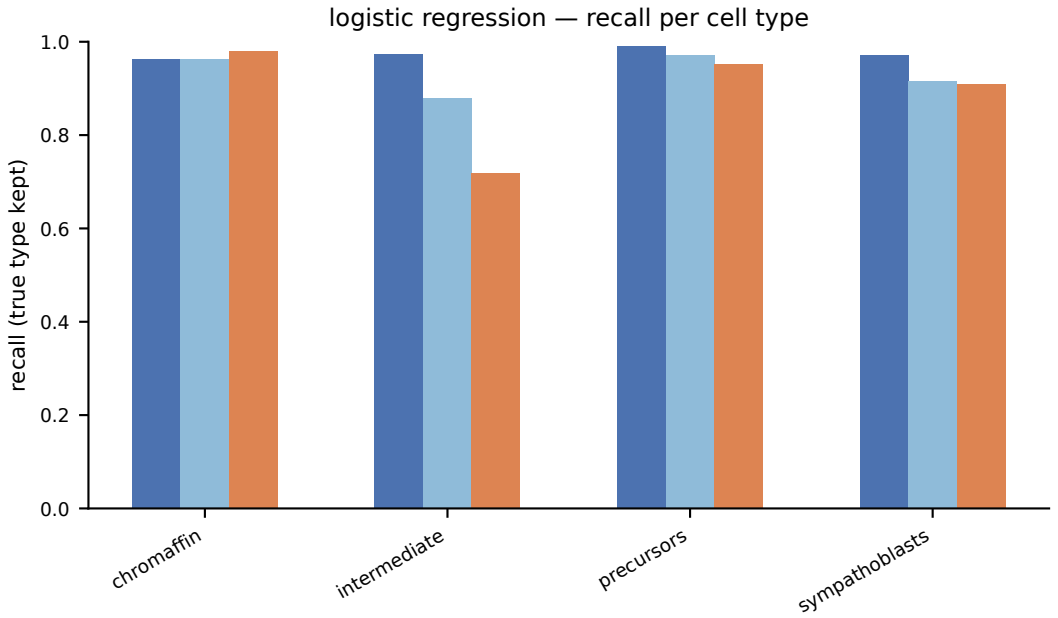
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.





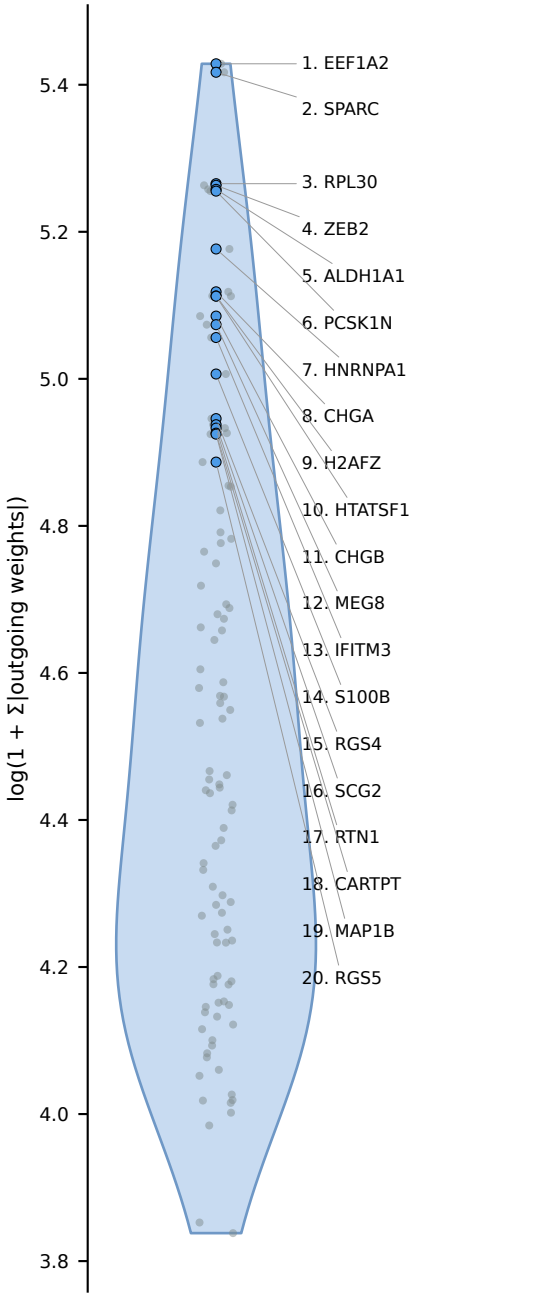




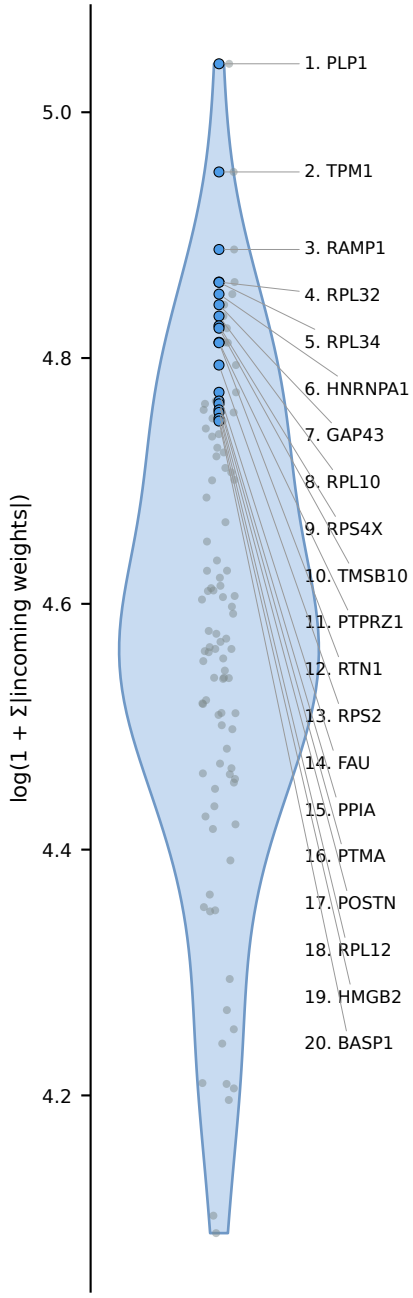
2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus not shown (stimulus = 0.2 < 0.5).

A



B



C Top 20 regulators

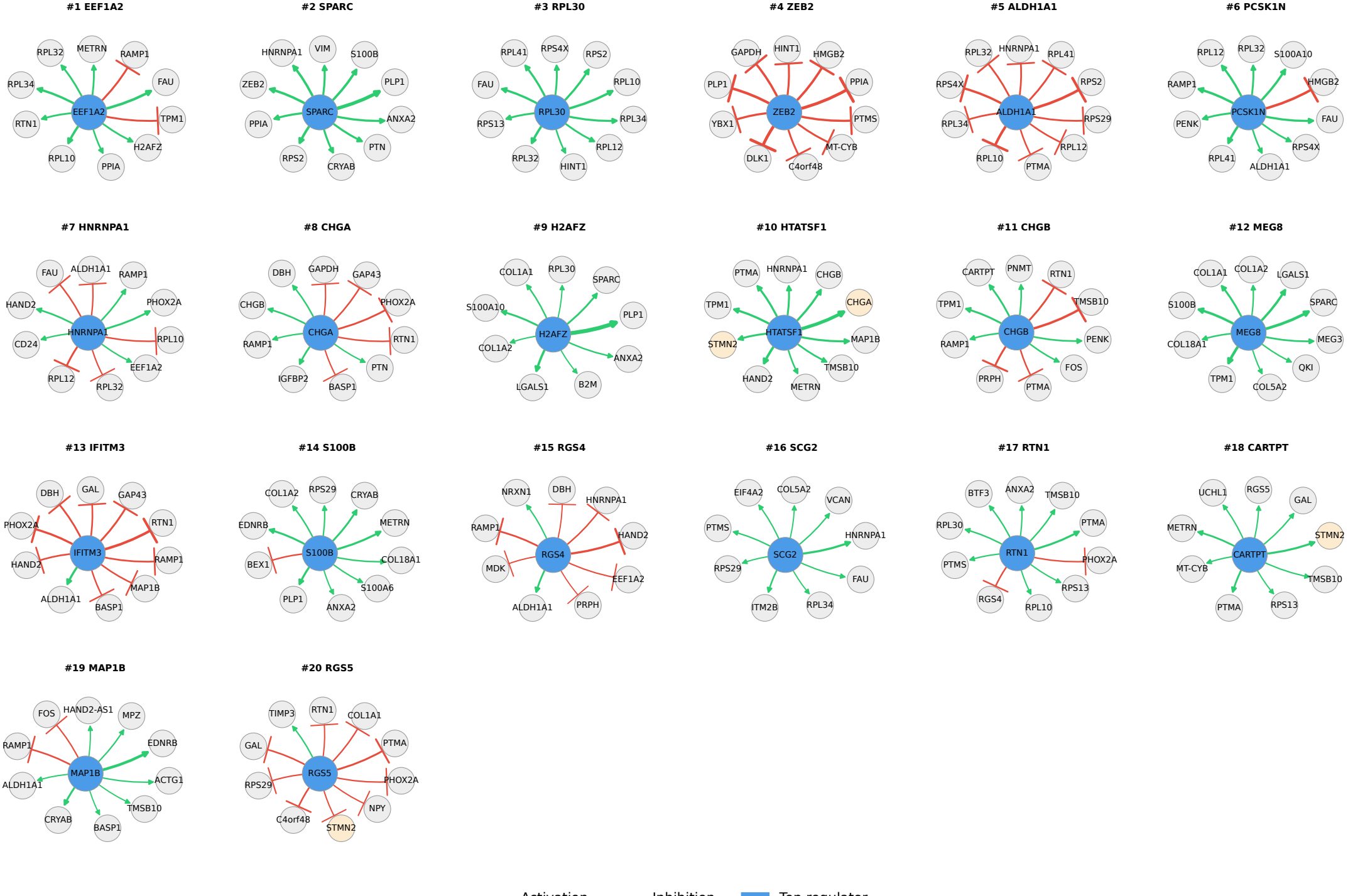
#	Regulator	Σ|w|	act.	inh.
1	EEF1A2	226.81	61	29
2	SPARC	224.20	53	29
3	RPL30	192.53	80	15
4	ZEB2	192.11	9	80
5	ALDH1A1	191.04	14	81
6	PCSK1N	190.58	62	31
7	HNRNPA1	176.08	37	56
8	CHGA	166.06	59	34
9	H2AFZ	165.13	68	25
10	HTATSF1	165.07	68	18
11	CHGB	160.63	52	37
12	MEG8	158.76	60	34
13	IFITM3	155.99	31	56
14	S100B	148.39	45	47
15	RGS4	139.60	50	42
16	SCG2	138.44	69	22
17	RTN1	137.79	64	25
18	CARTPT	136.85	77	15
19	MAP1B	136.66	53	42
20	RGS5	131.52	38	54

D Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	PLP1	153.38	61	29
2	TPM1	140.38	47	47
3	RAMP1	131.73	46	46
4	RPL32	128.26	53	42
5	RPL34	128.23	49	42
6	HNRNPA1	127.00	49	40
7	GAP43	125.90	41	50
8	RPL10	124.73	51	44
9	RPS4X	123.74	44	47
10	TMSB10	123.50	45	47
11	PTPRZ1	122.08	58	35
12	RTN1	122.03	48	45
13	RPS2	119.82	50	45
14	FAU	117.17	40	52
15	PPIA	116.33	44	49
16	PTMA	116.07	47	47
17	POSTN	115.49	56	30
18	RPL12	115.25	53	40
19	HMGB2	114.69	38	51
20	BASP1	114.44	41	51

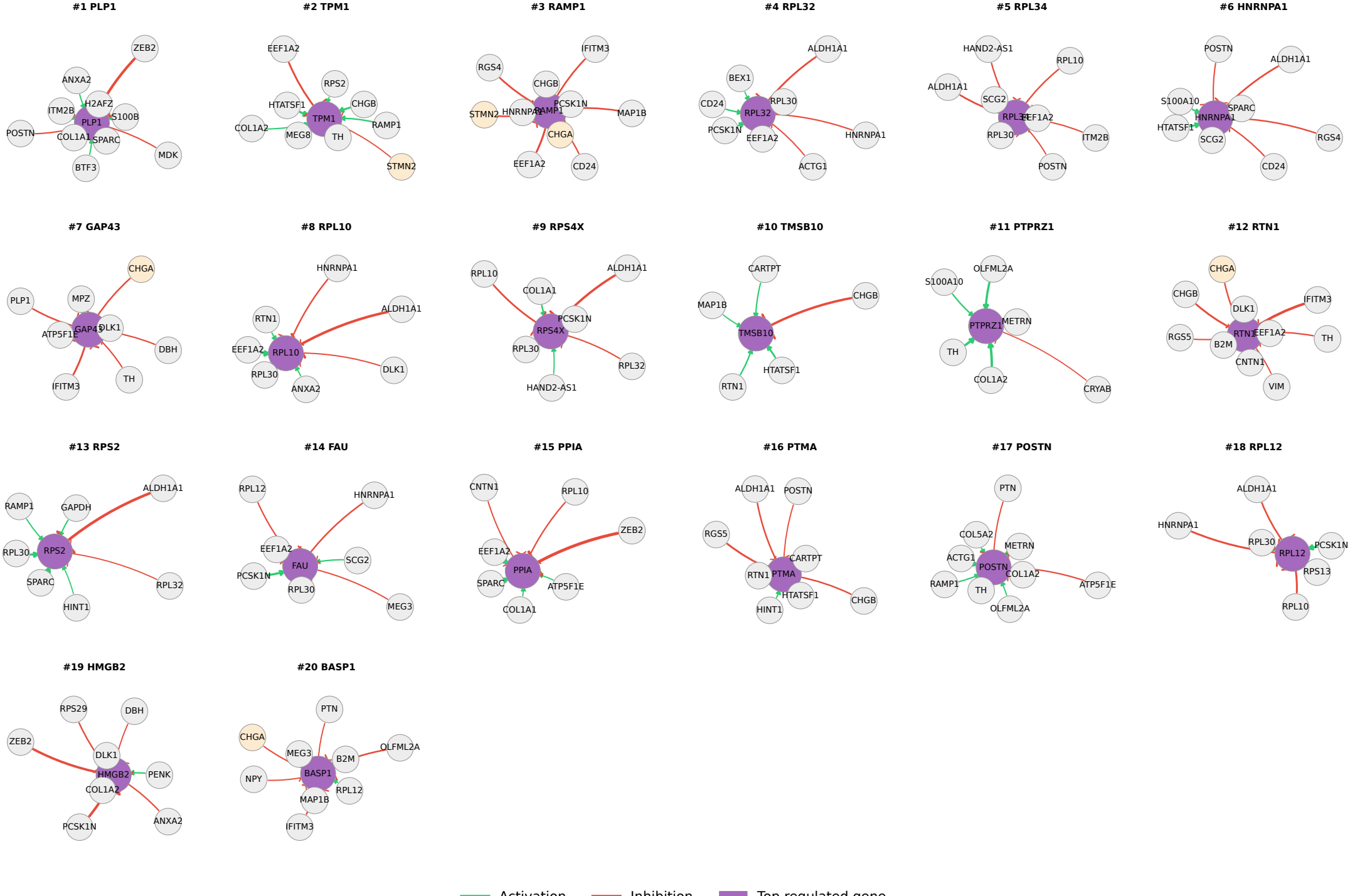
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



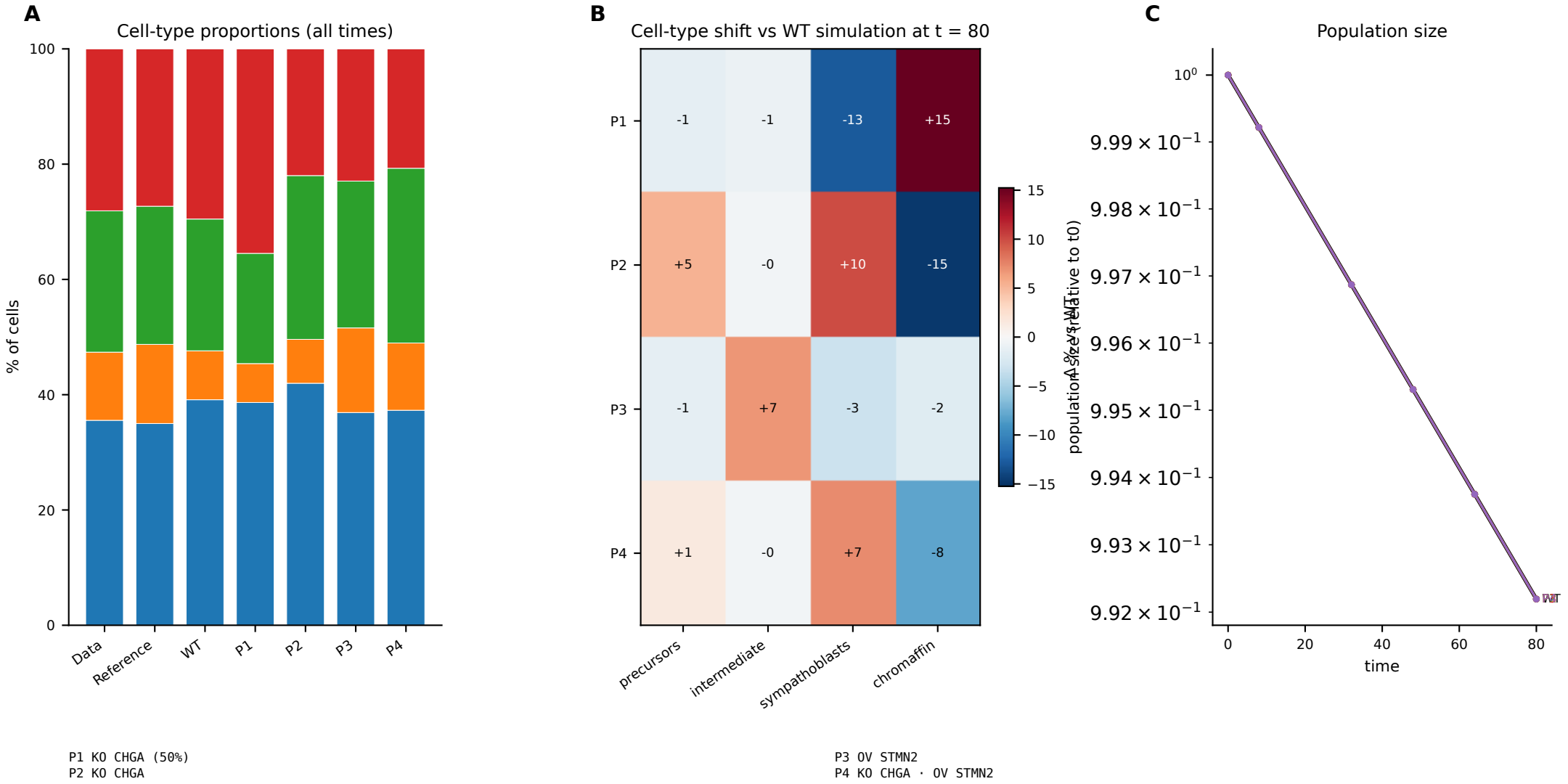
2. Gene regulatory network — top regulated genes and their main regulators

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



3. In-silico perturbations — overview

Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (proliferation MLP + RATE effects).

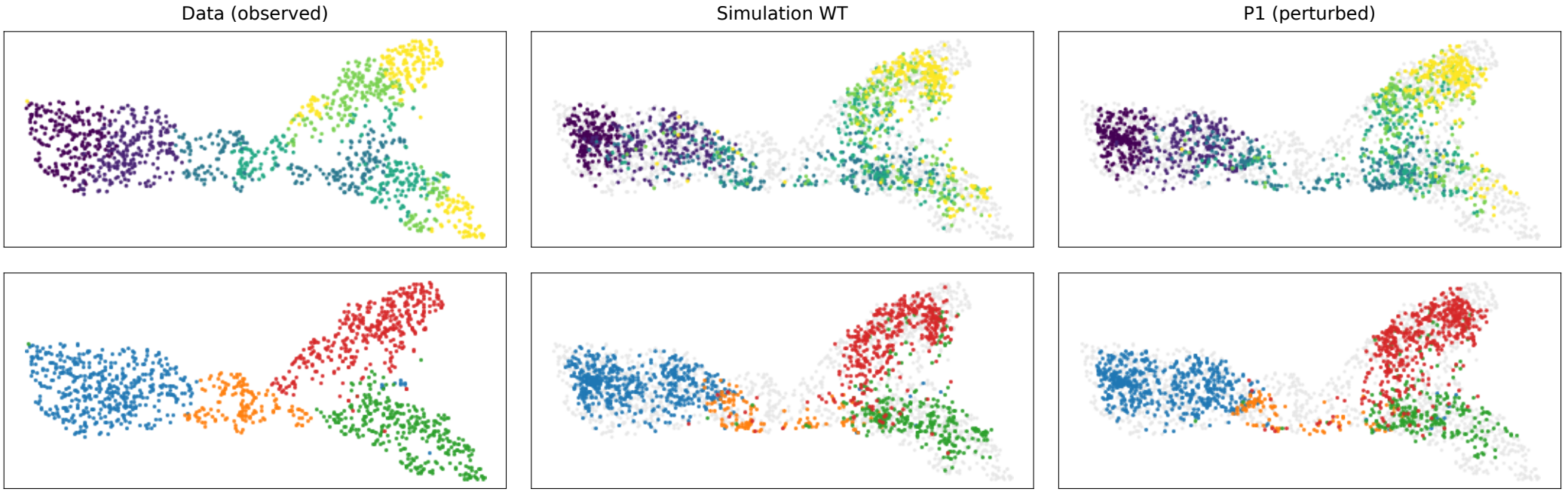


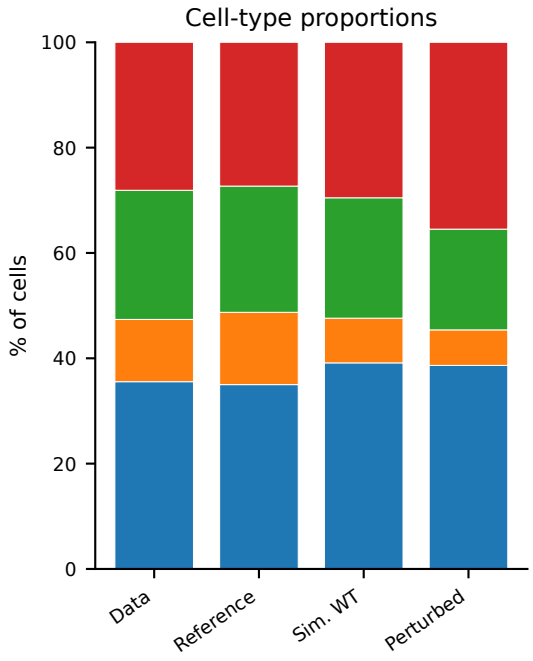
P1 KO CHGA (50%)
P2 KO CHGA

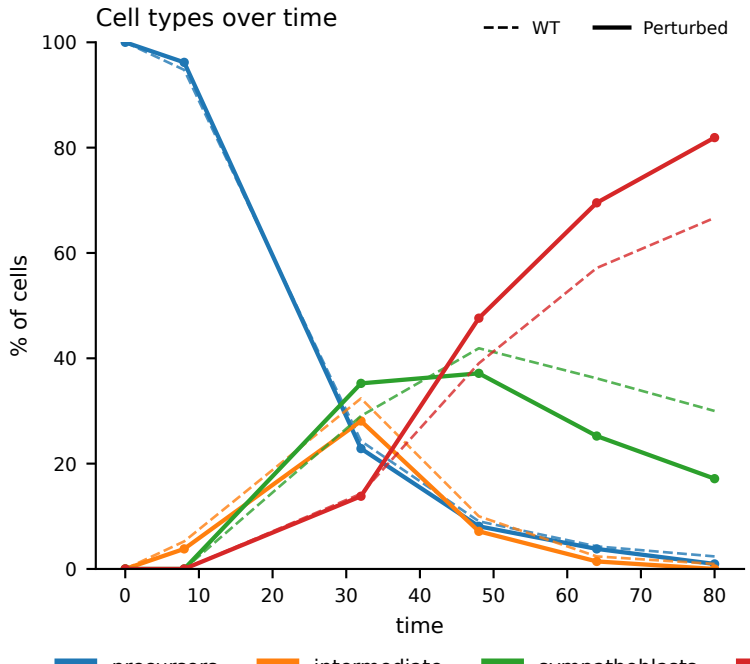
P3 OV STMN2
P4 KO CHGA · OV STMN2

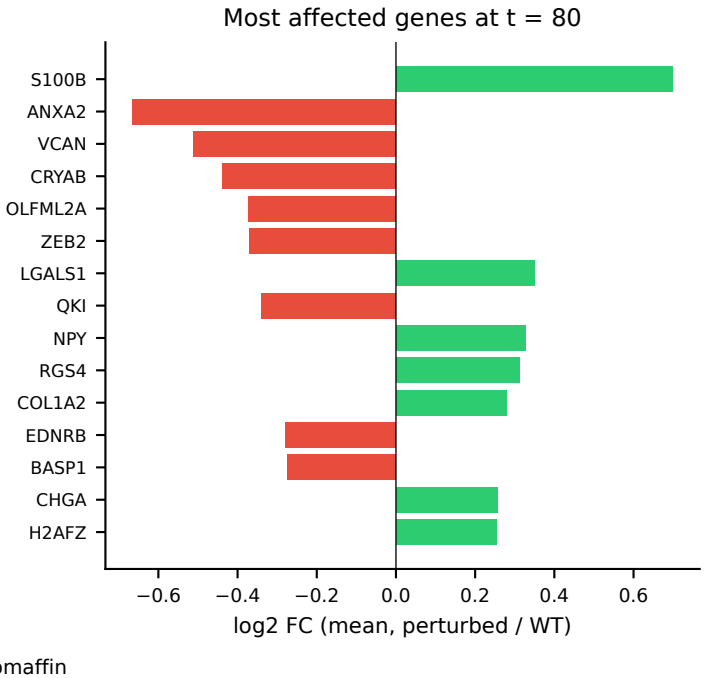
3. Perturbation P1 — KO CHGA (50%)

KO_CHGApct50_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 80: ×1.00 vs WT.



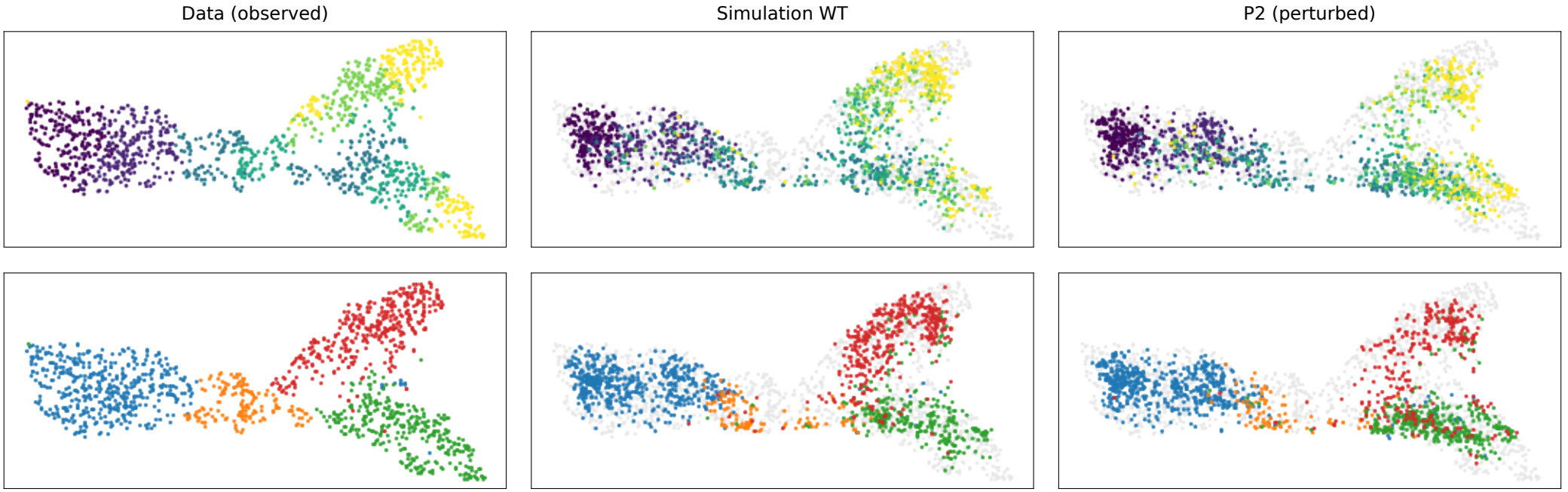


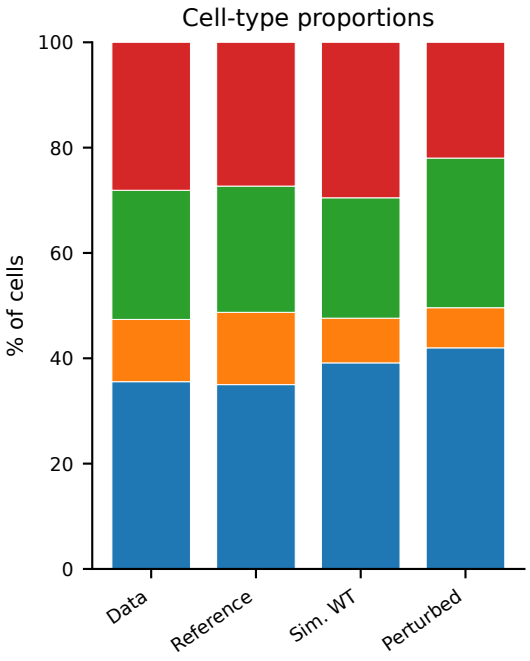


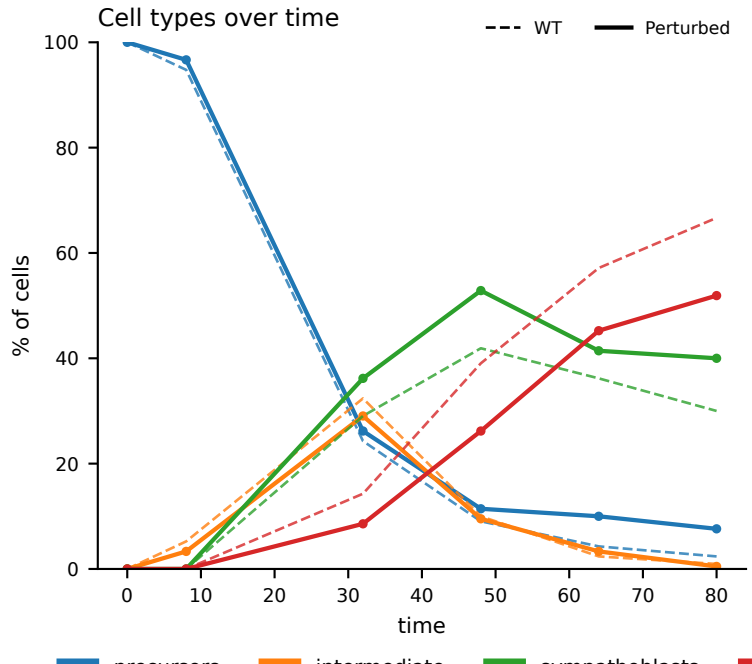


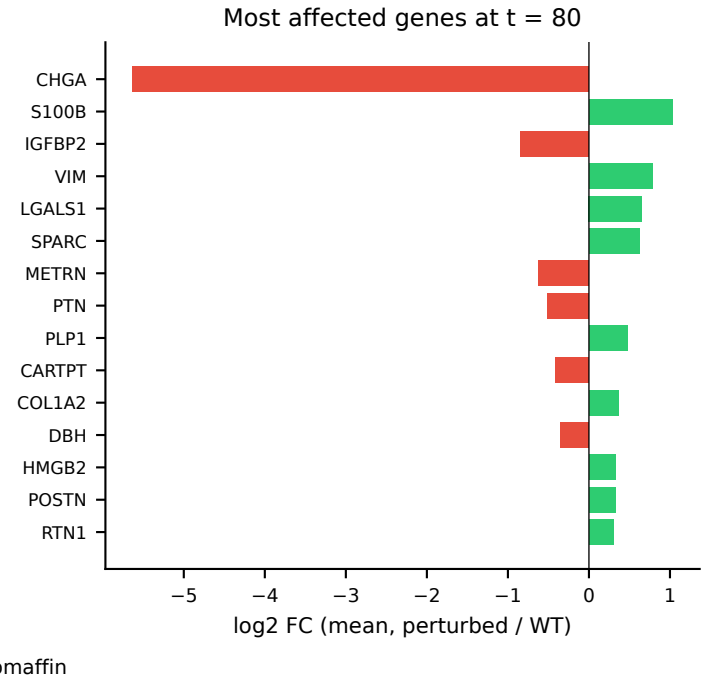
3. Perturbation P2 — KO CHGA

KO_CHGA_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 80: ×1.00 vs WT.



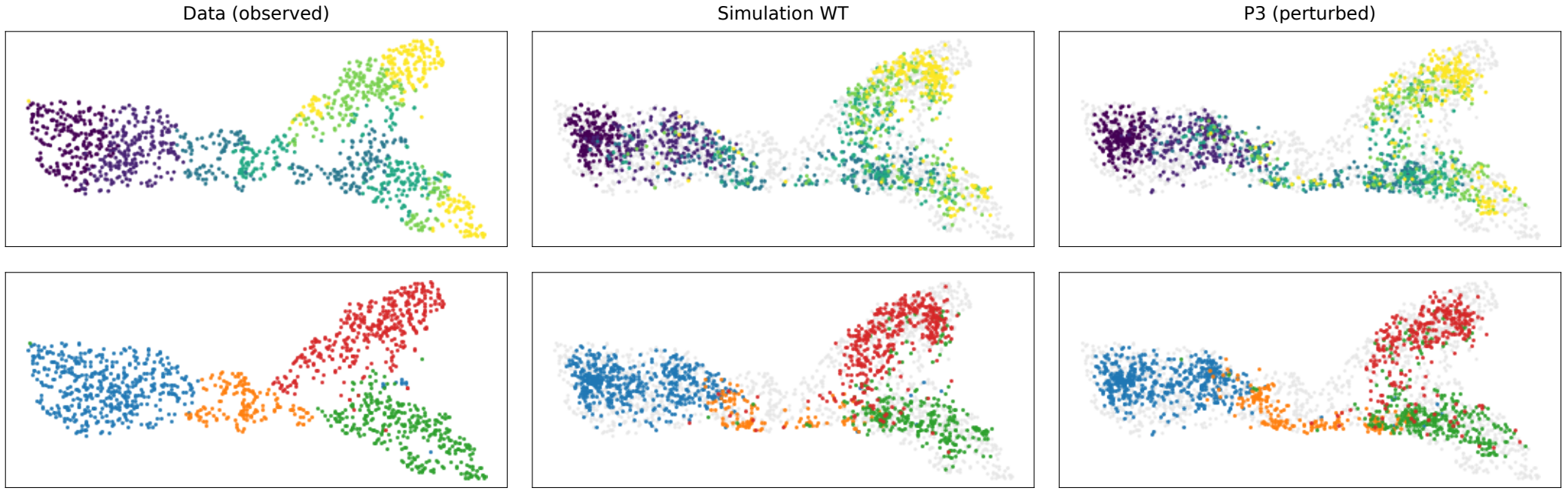


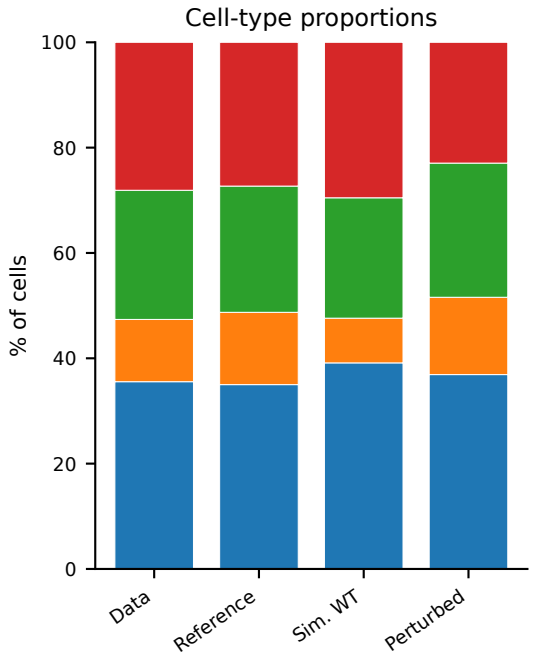


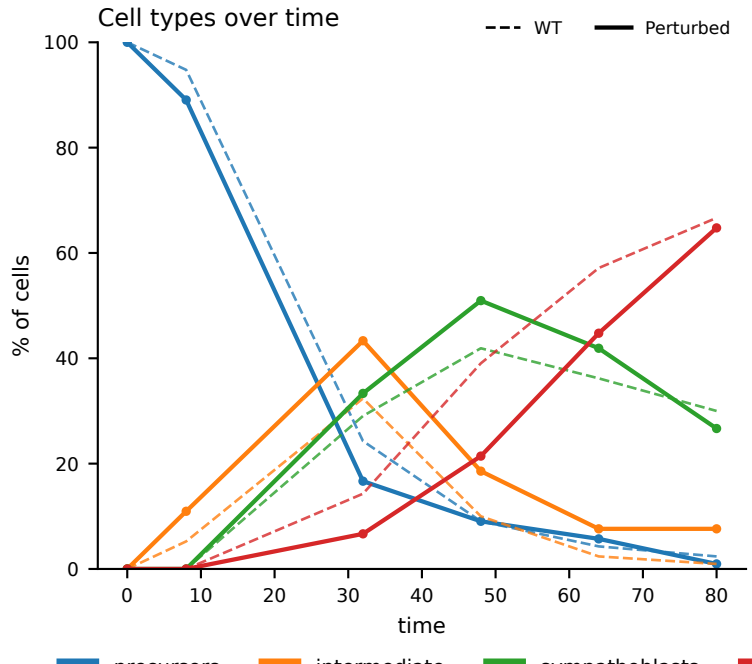


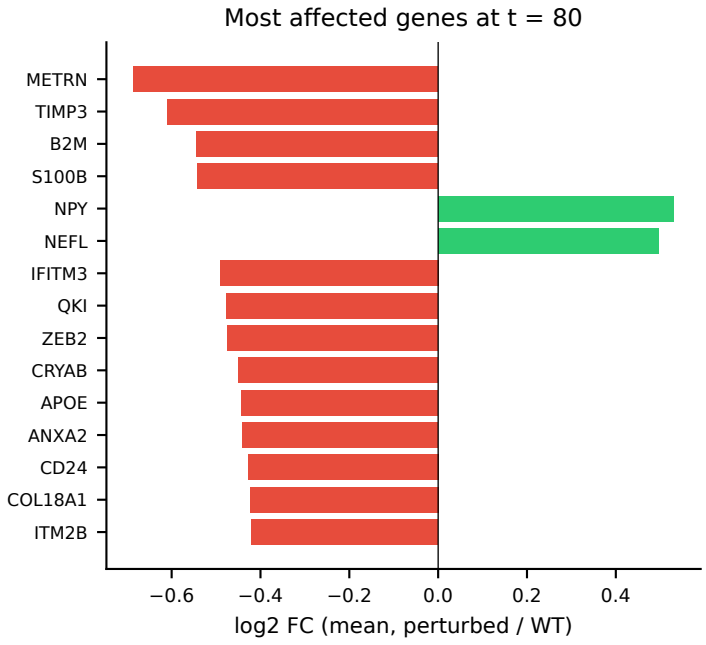
3. Perturbation P3 — OV STMN2

KO_none_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 80: ×1.00 vs WT.



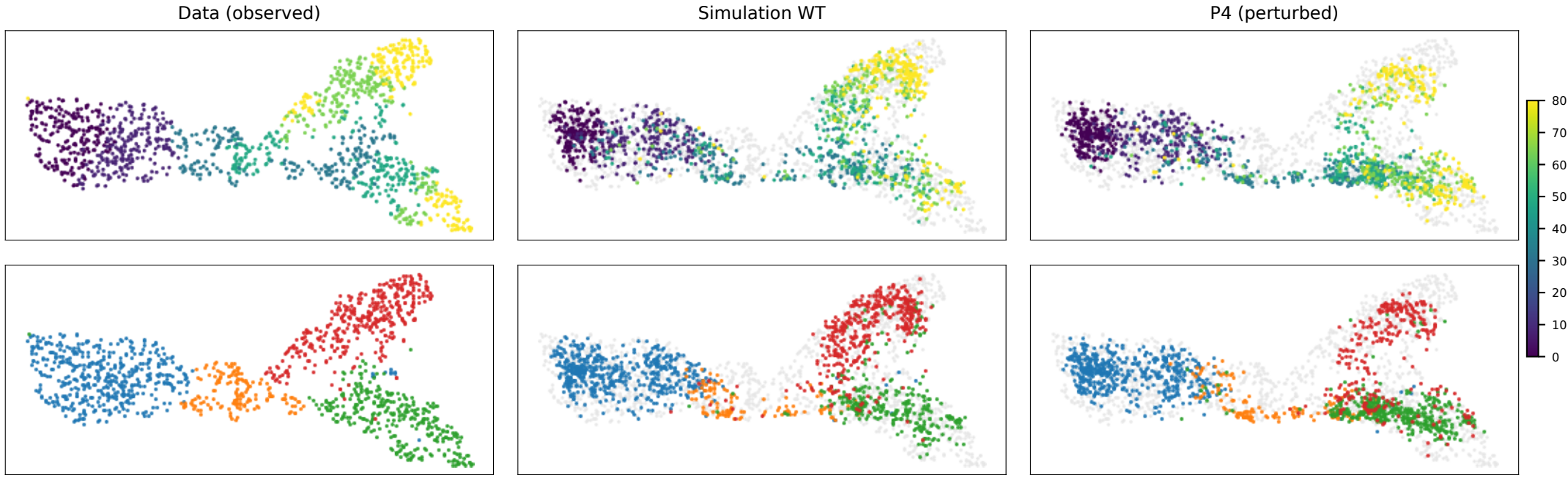


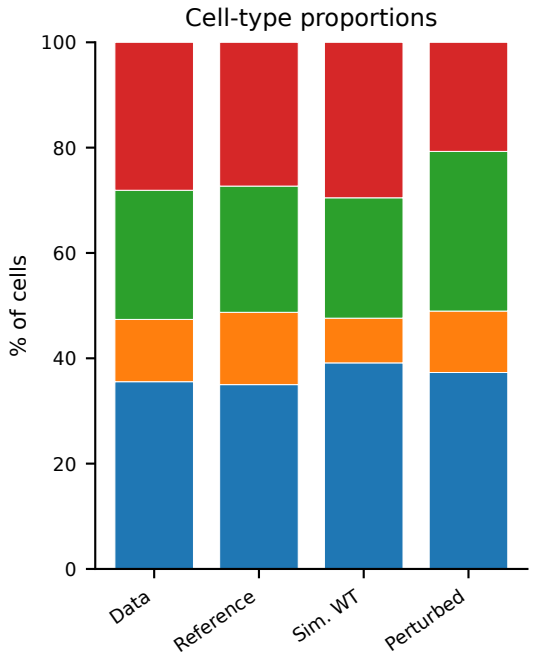


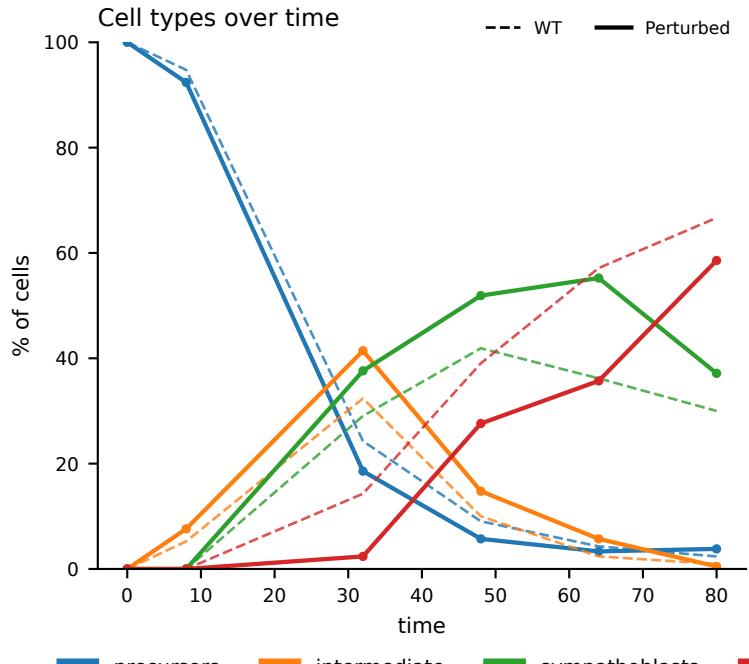


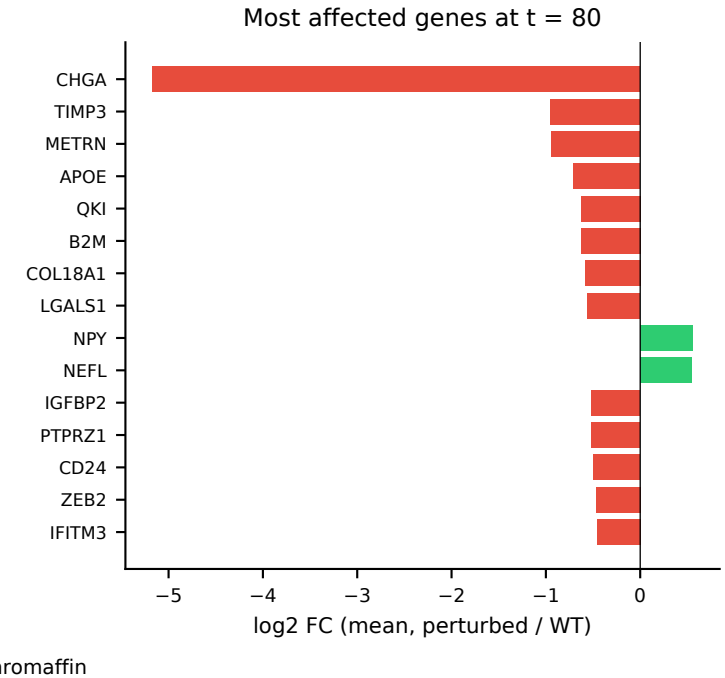
3. Perturbation P4 — KO CHGA · OV STMN2

KO_CHGA_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data). Population at t = 80: ×1.00 vs WT.









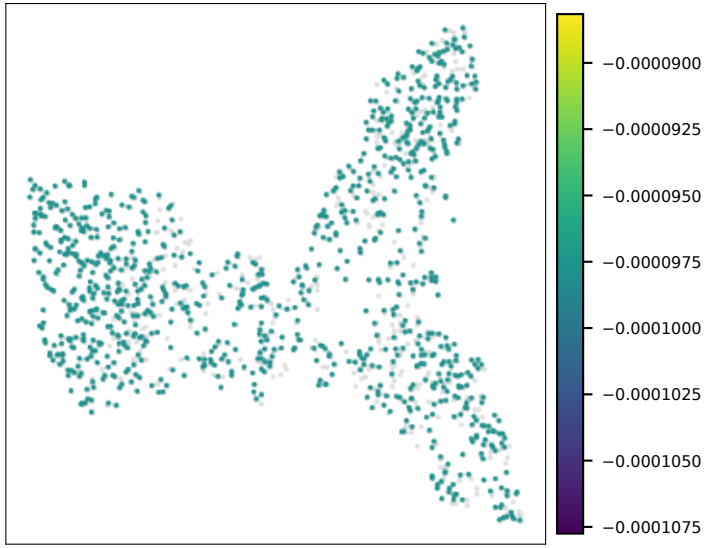


4. Proliferation — prior vs learned net rate (birth − death)

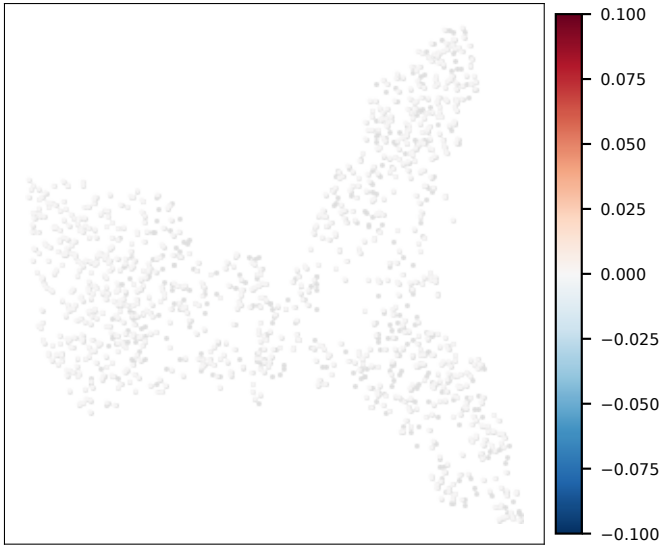
Learned = MLP R(P), mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)

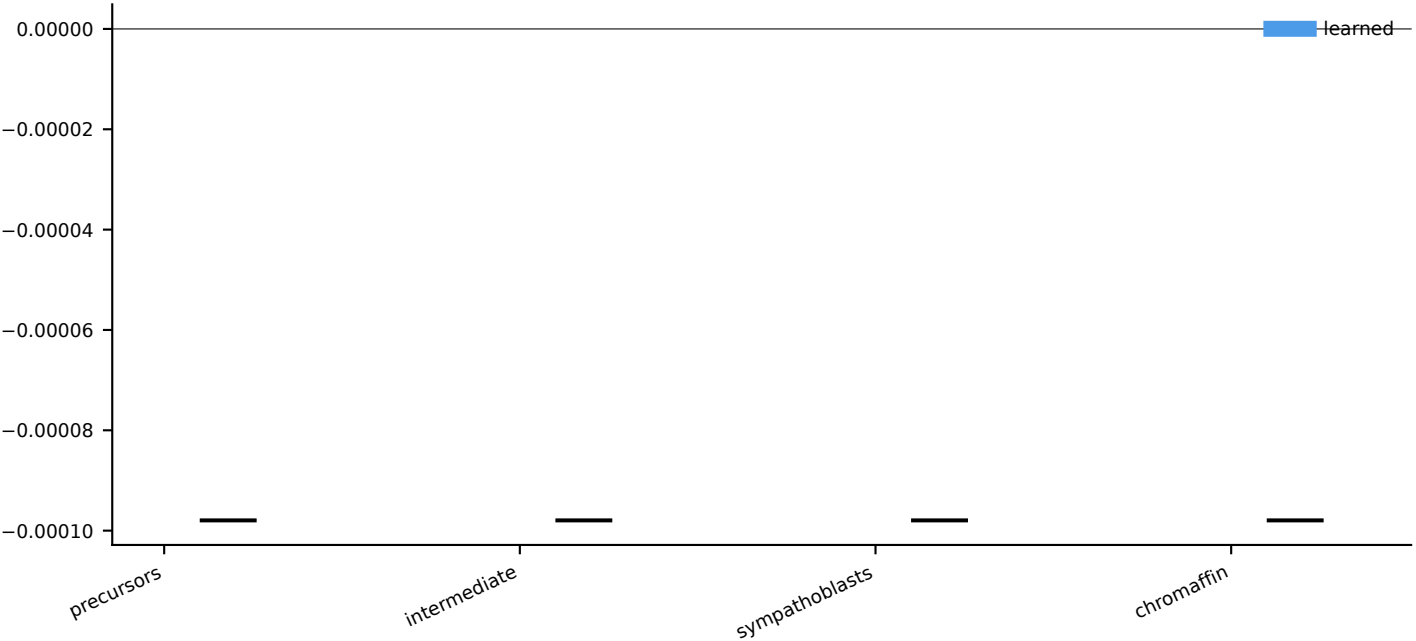
Learned net rate (MLP R(P))



Learned − mean



Net proliferation rate per cell type (cell_type)

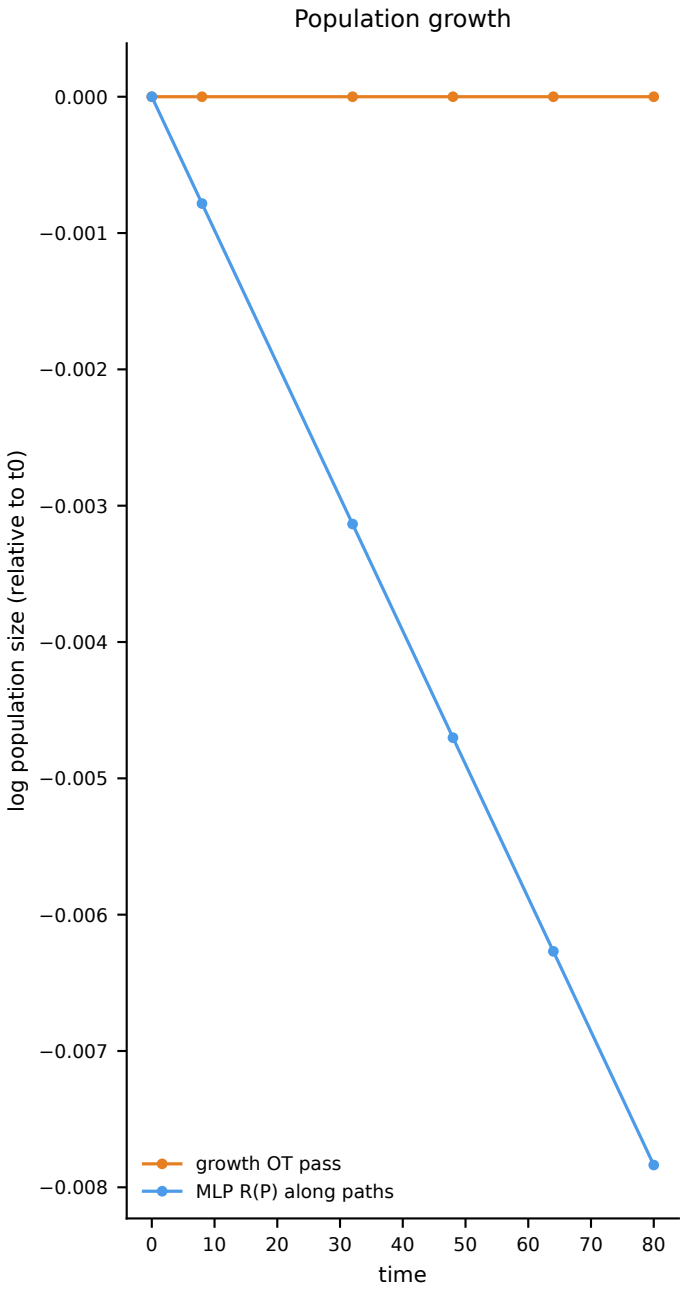


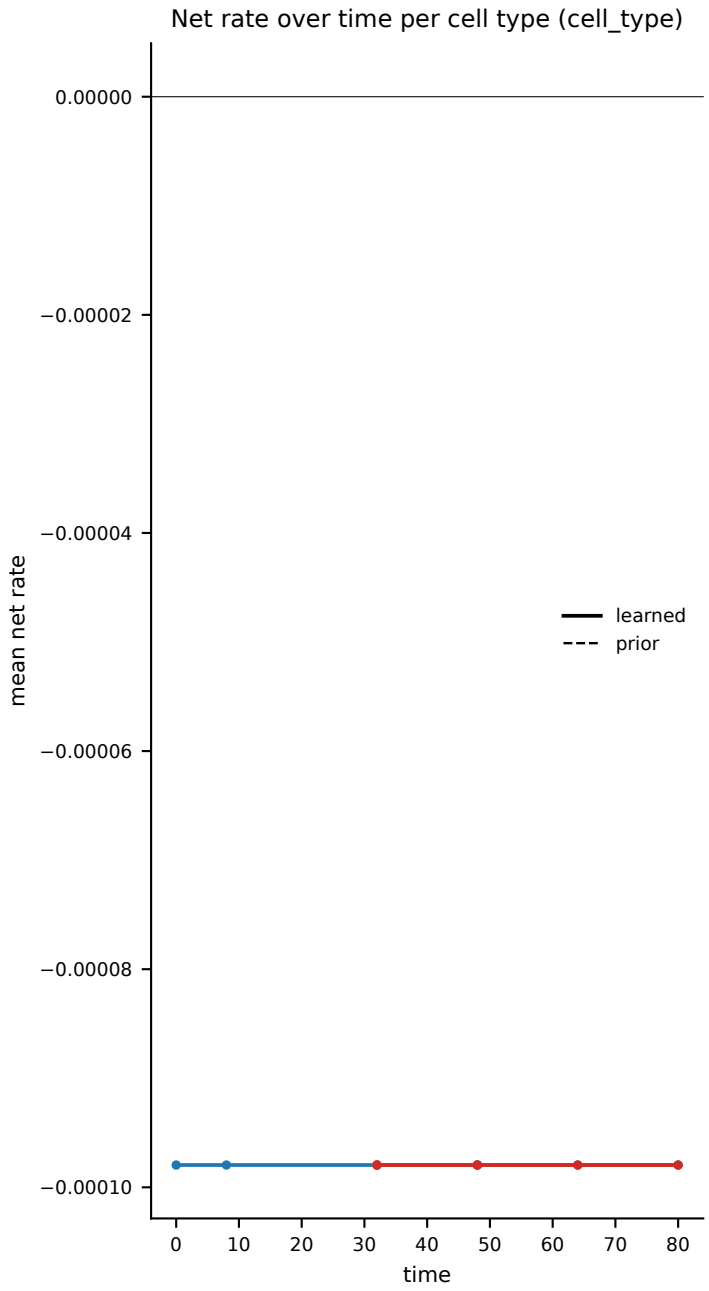
Mean net rate per cell type (cell_type)

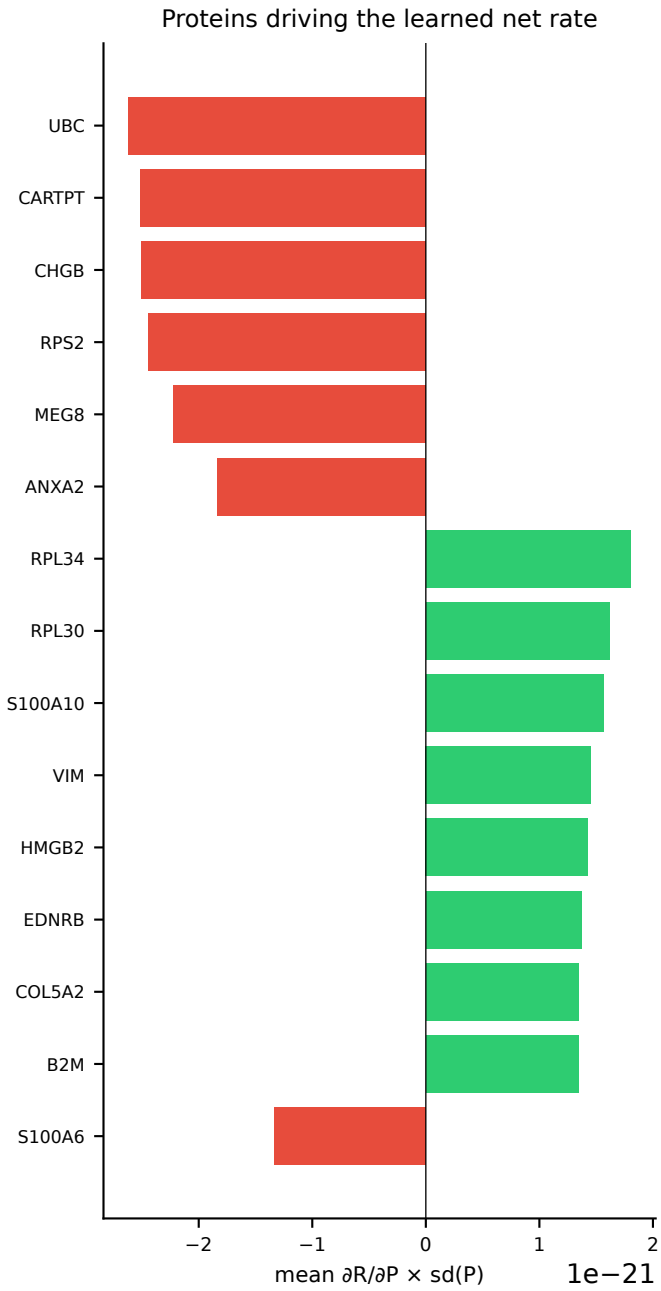
Cell type	Anchor	Prior	Learned
precursors	—	—	-9.8e-05
intermediate	—	—	-9.8e-05
sympathoblasts	—	—	-9.8e-05
chromaffin	—	—	-9.8e-05

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; its absolute level follows the prior (anchored on the sheet population_sizes by fit_population_anchors, if filled).



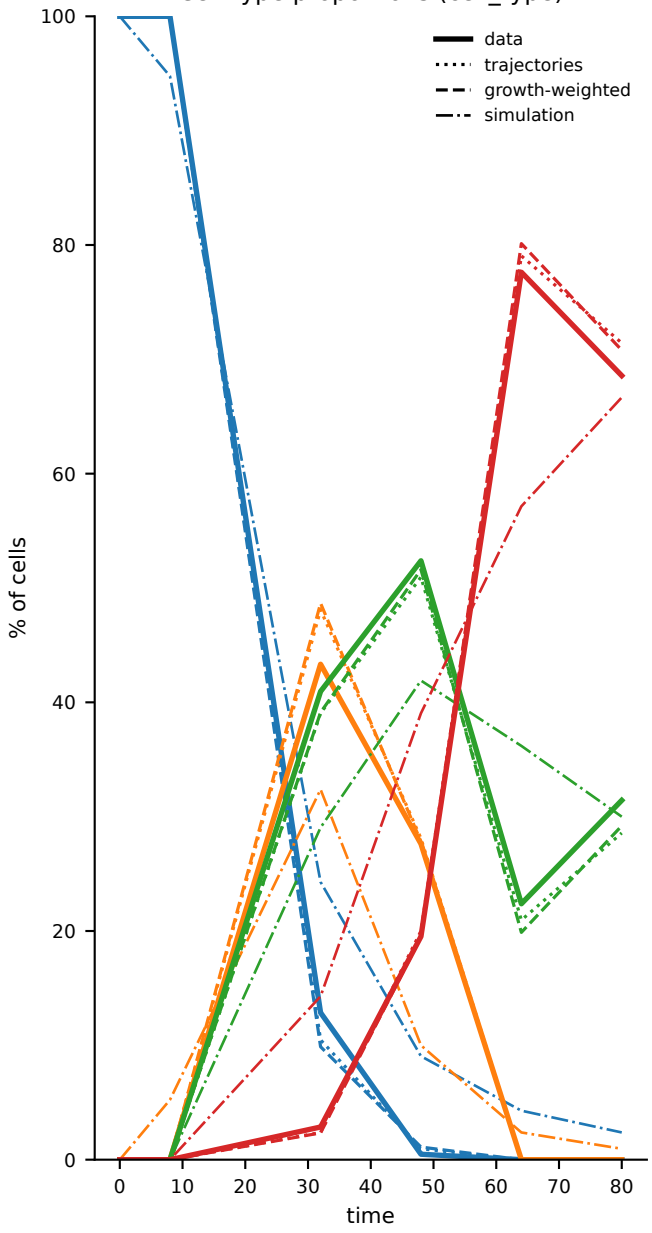


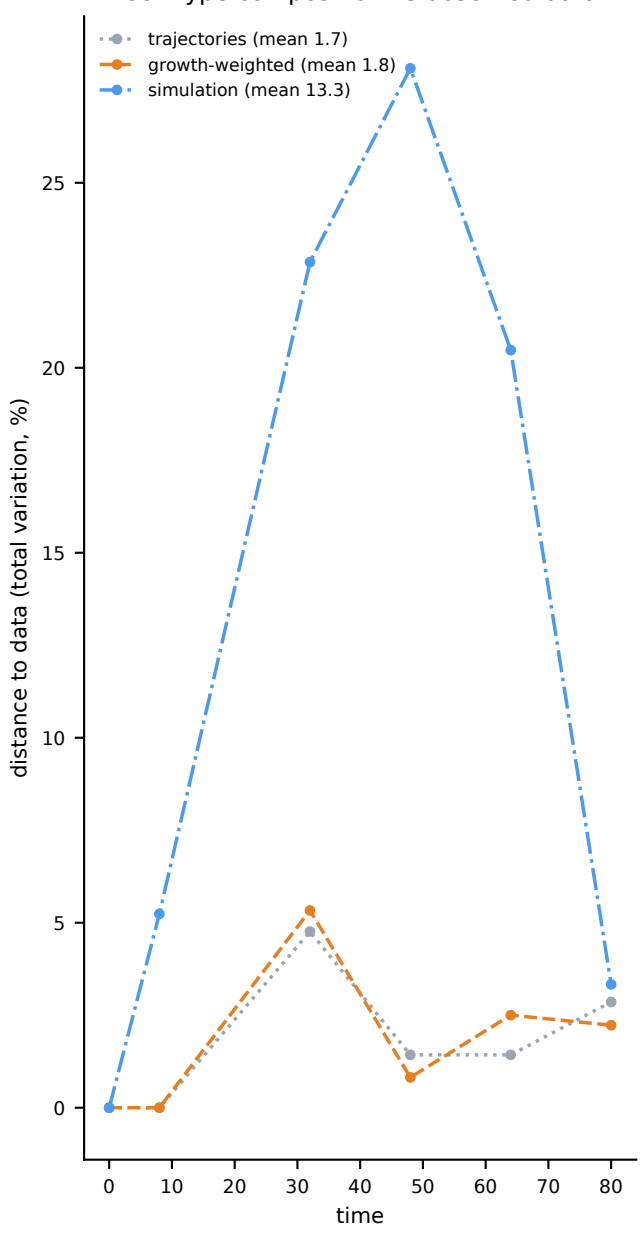


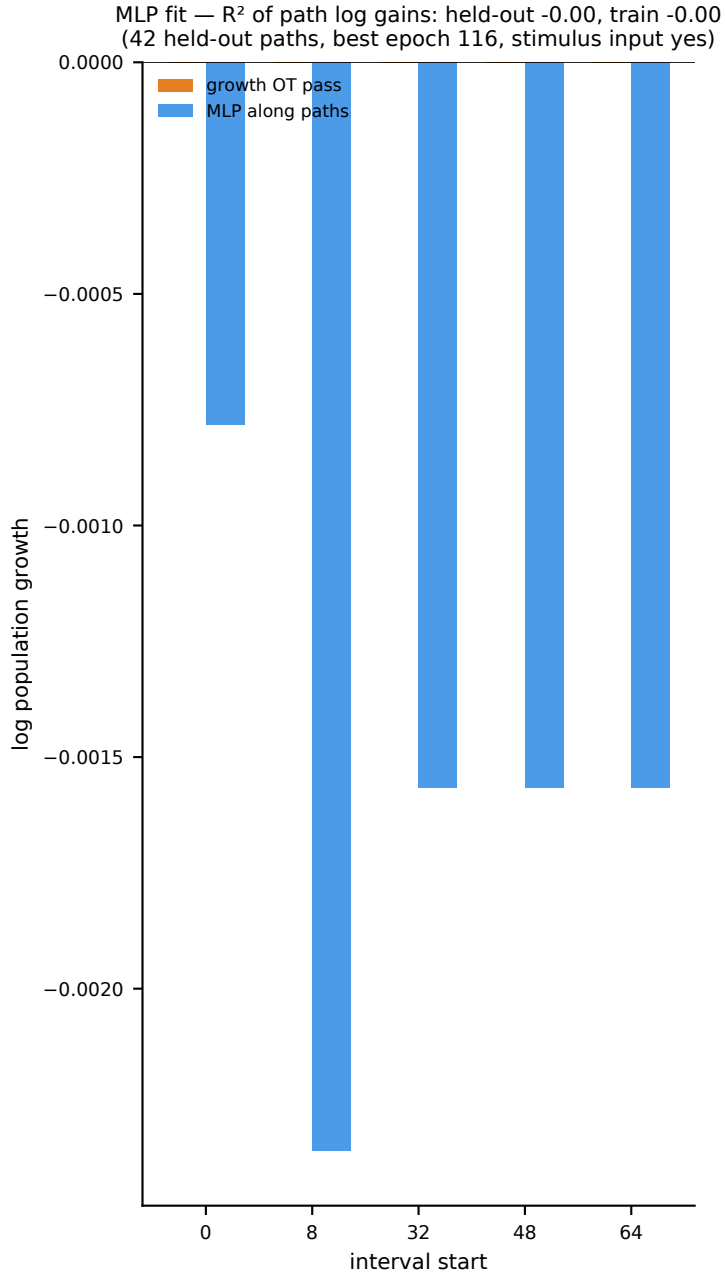
precursors · intermediate · sympathoblasts · chromaffin

4. Proliferation — growth-weighted trajectories and MLP fit

The OT selection keeps one descendant per ancestor (trajectories without proliferation); weighting each path by its expansion exp(∫R_opt) should bring them back to the observed data. The simulation used the proliferation MLP.









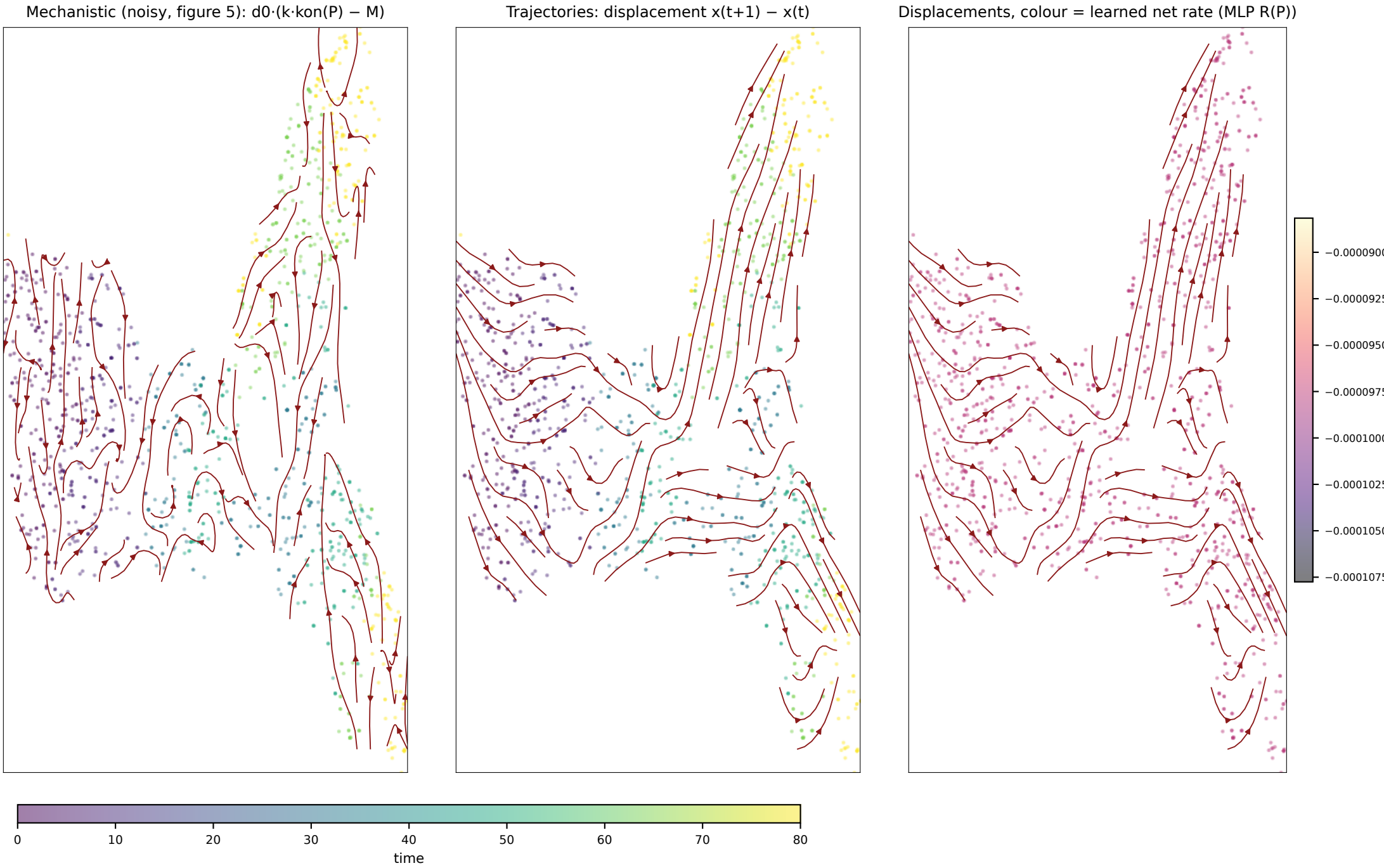
4. Protein dilution

Section unavailable:

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

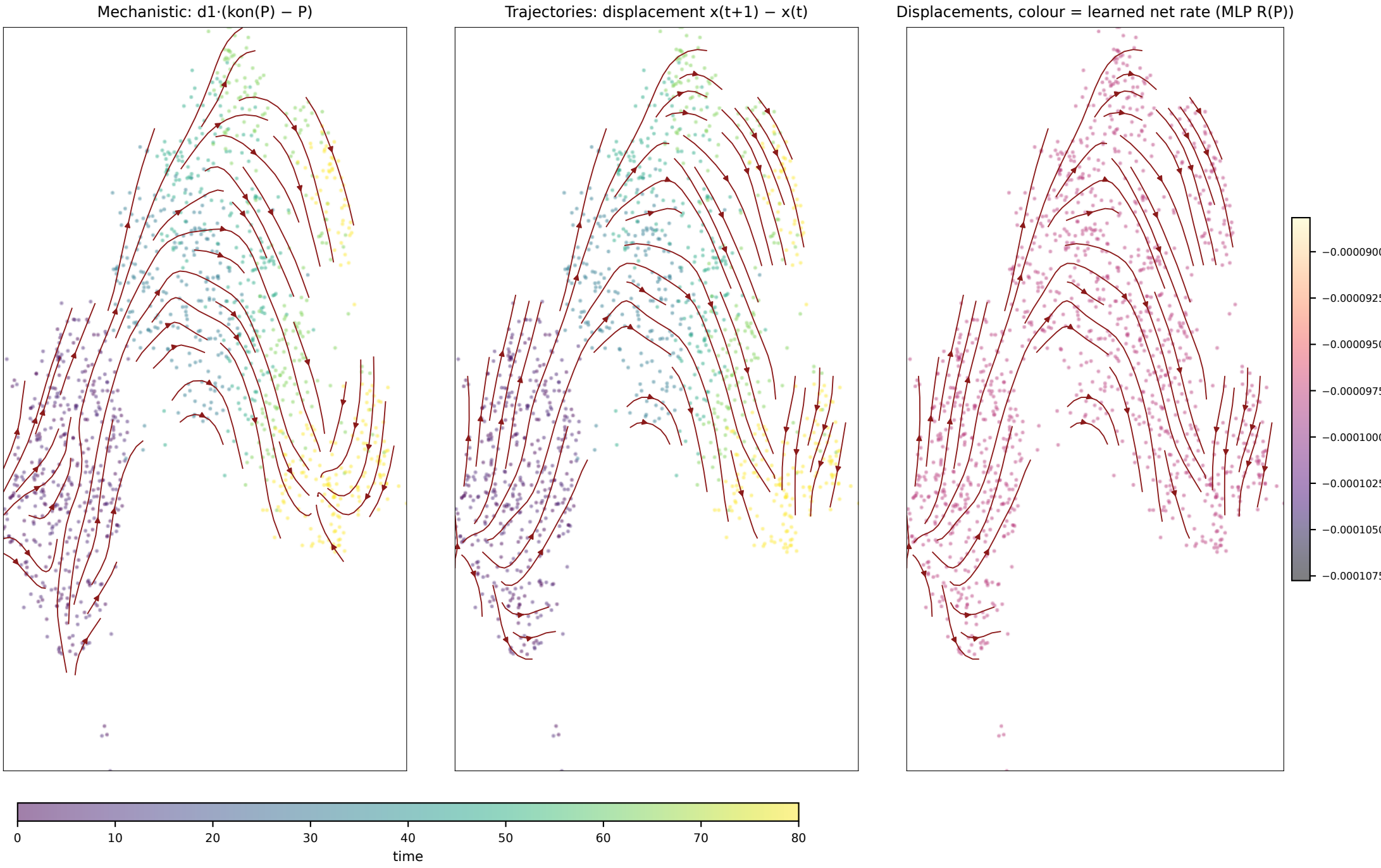
5. Learned dynamics — mRNA velocity and displacement fields

All 1260 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.03.



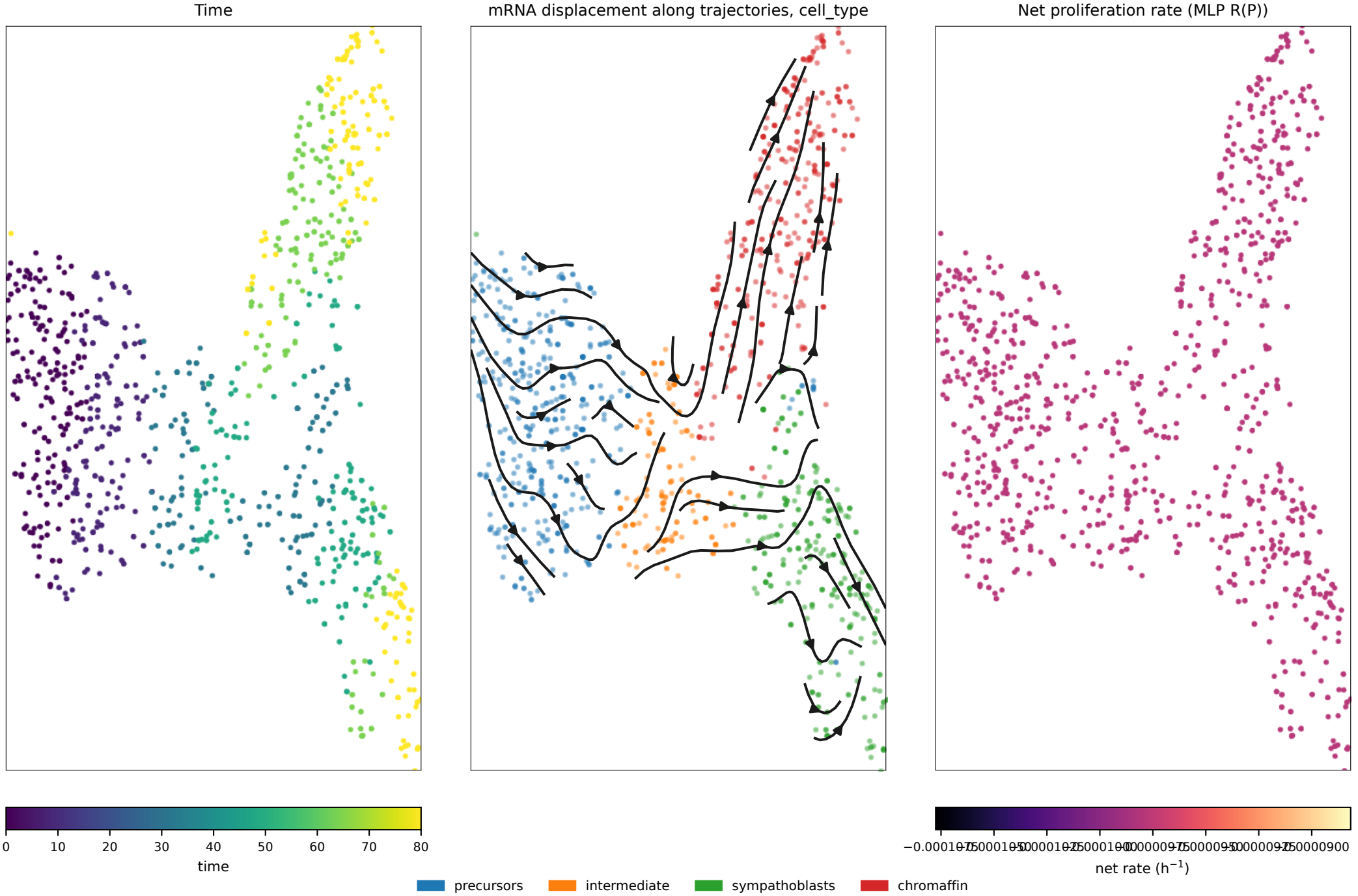
5. Learned dynamics — Proteins velocity and displacement fields

All 1260 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.57.



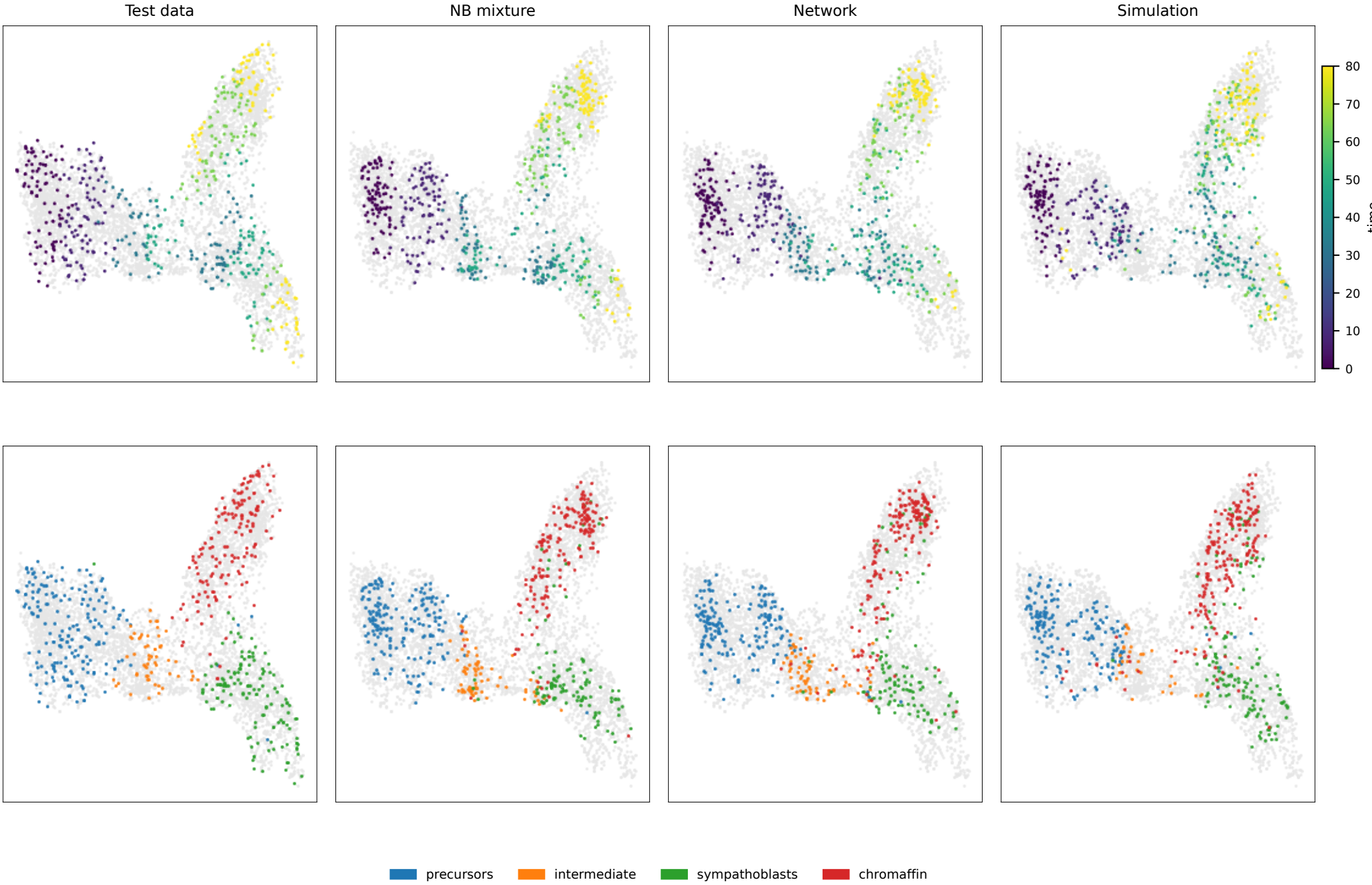
5. Learned dynamics — summary

All 1260 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

